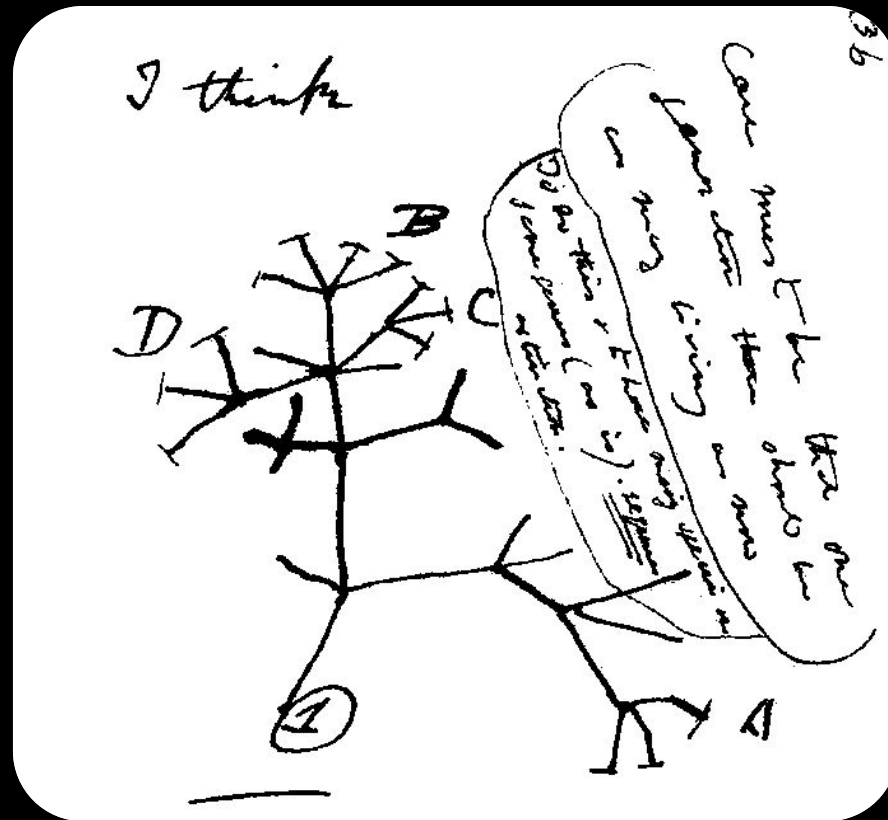




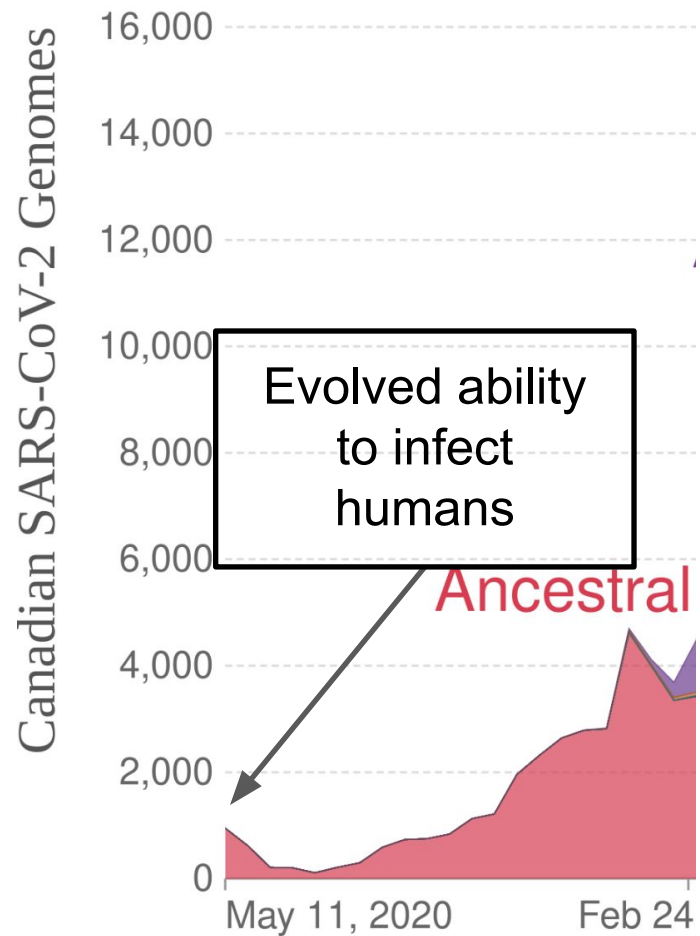
# Sequence change and evolution

# Overview

1. Mutations are changes that are passed from generation to generation
2. Many different types of mutation are possible, each can have different impacts on encoded proteins
3. Mutations can have consequences for the organism, leading to differential fitness, selection, and speciation

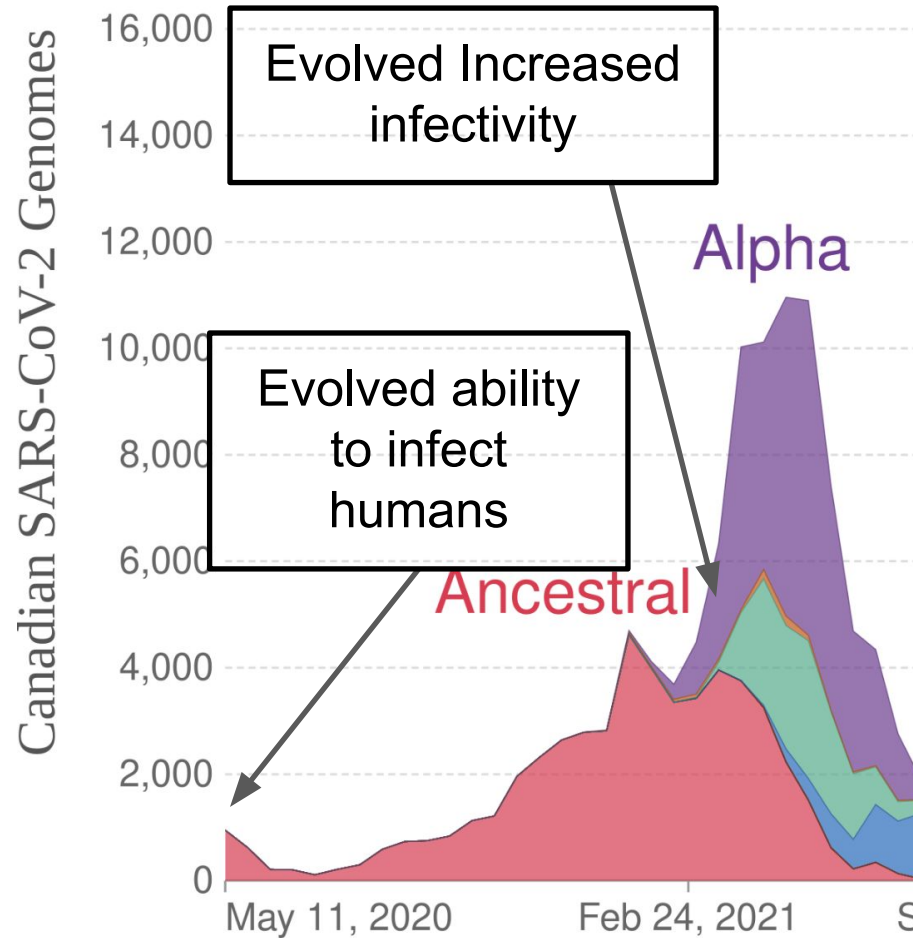
Why do we care about evolution?

# Evolution drives cases: SARS-CoV-2 waves



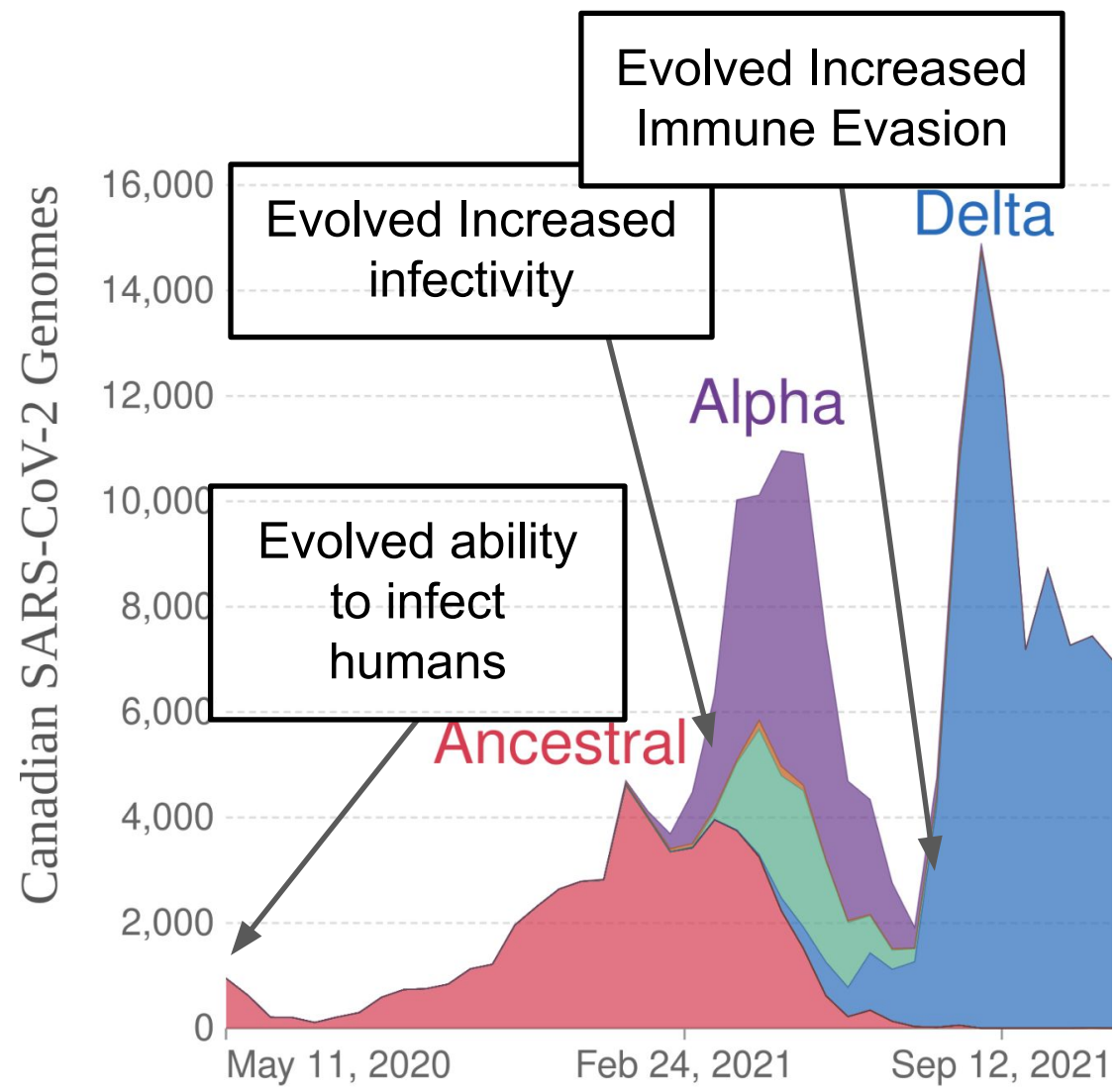
Source: GISAID, via CoVariants.org – Last up

# Evolution drives cases: SARS-CoV-2 waves



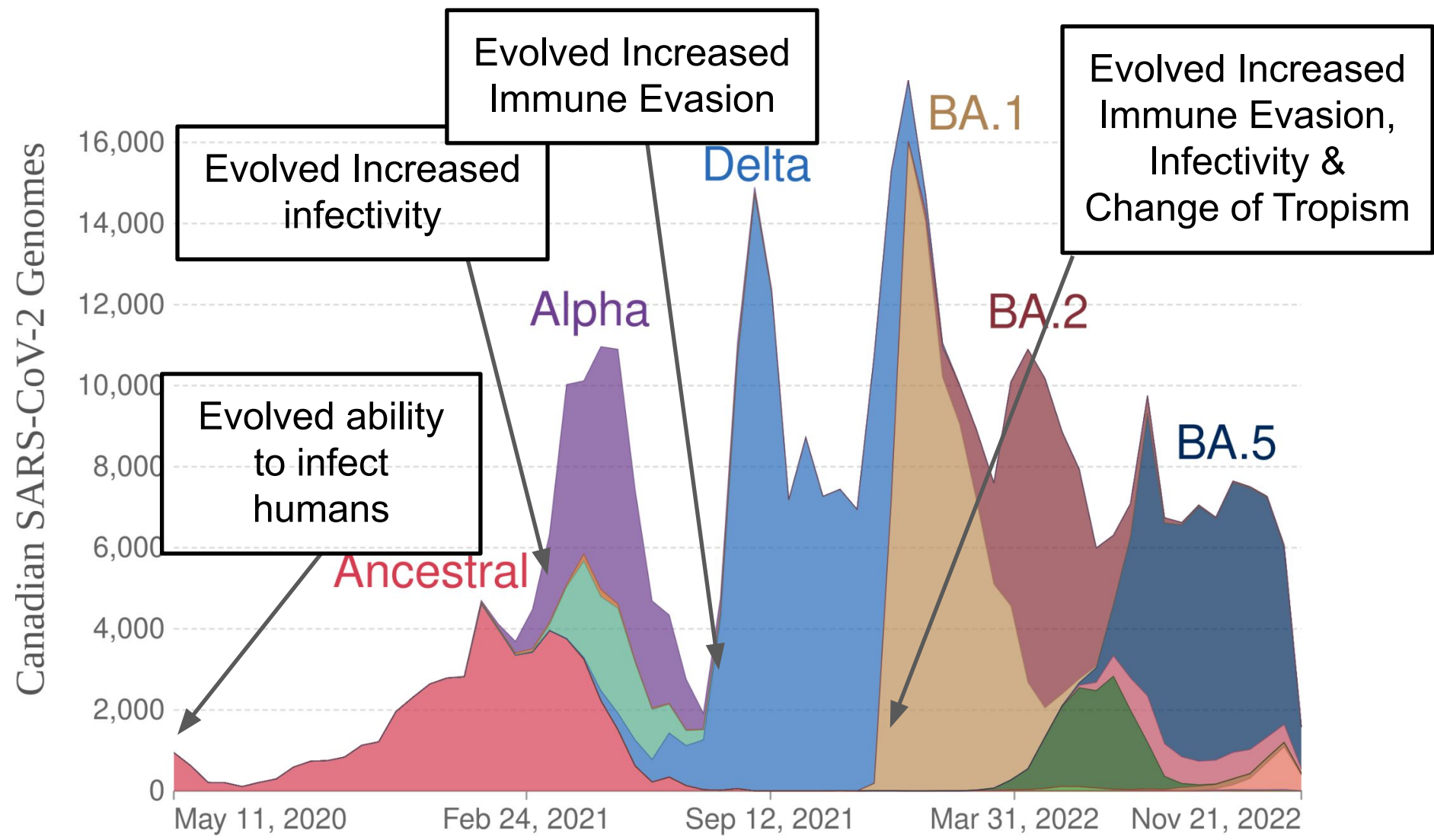
Source: GISAID, via CoVariants.org – Last updated 1 December

# Evolution drives cases: SARS-CoV-2 waves



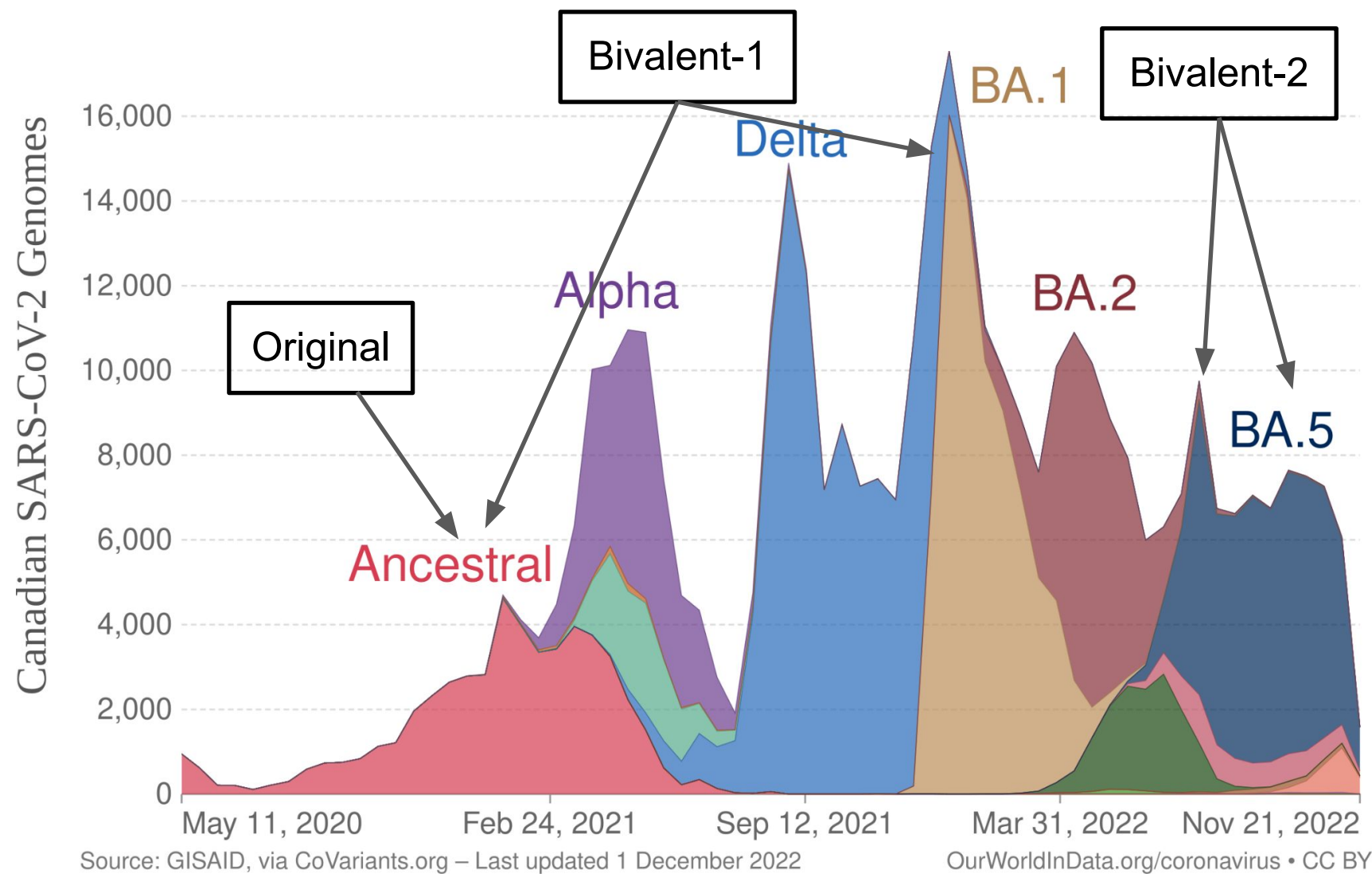
Source: GISAID, via CoVariants.org – Last updated 1 December 2022

# Evolution drives cases: SARS-CoV-2 waves



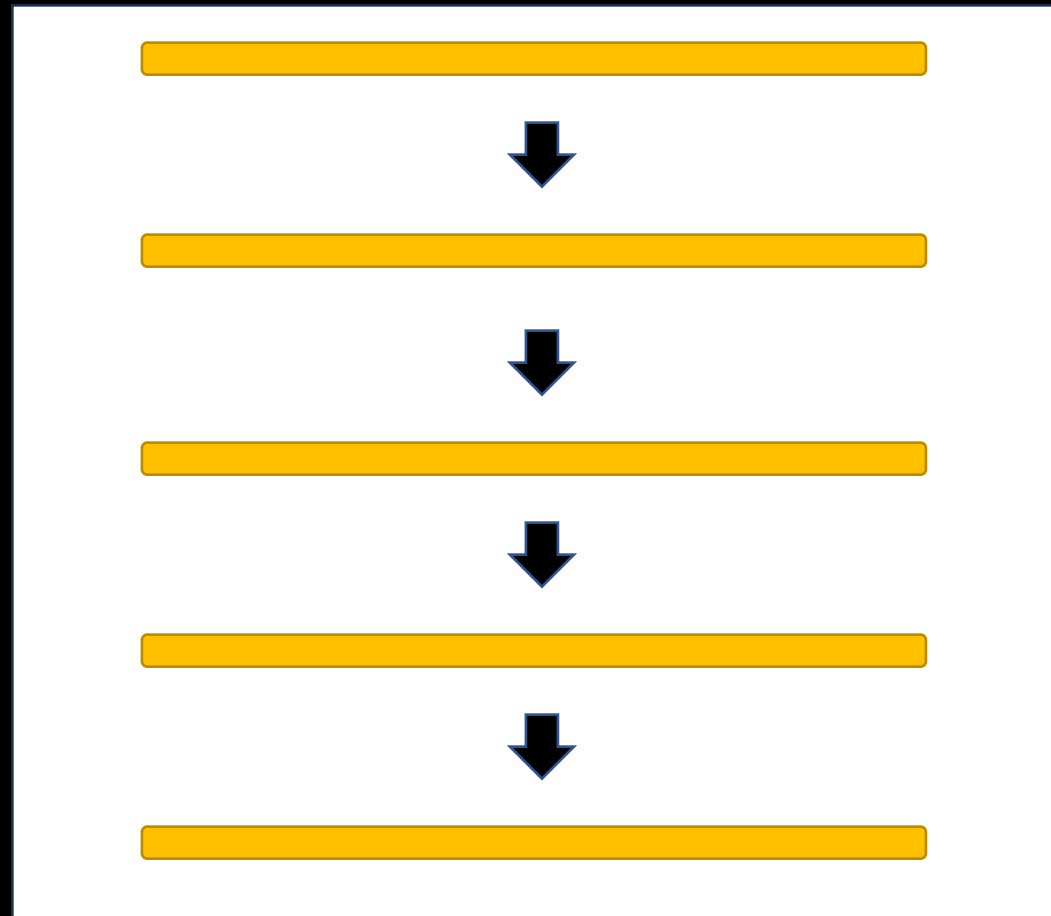
Source: GISAID, via CoVariants.org – Last updated 1 December 2022  
OurWorldInData.org/coronavirus • CC BY

# Evolution drives vaccine design/effectiveness

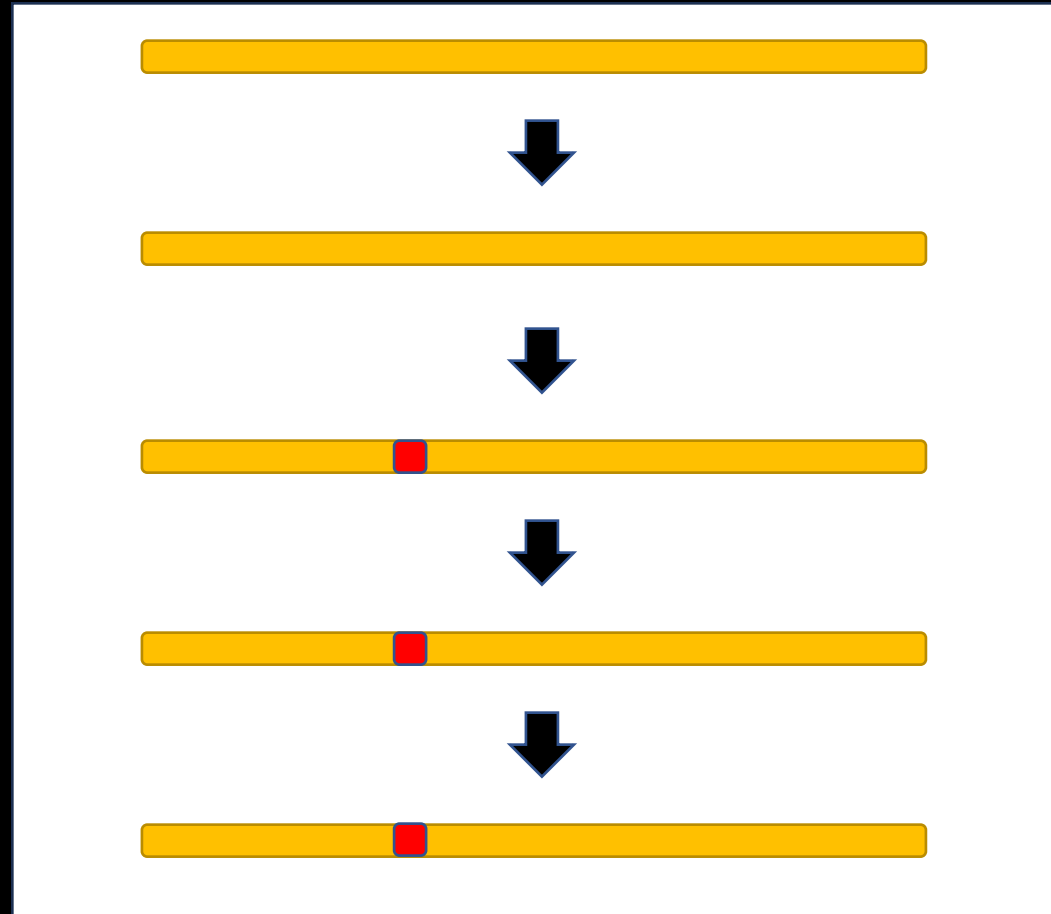




DNA is passed from parent to offspring  
through *replication*



Replication is not always 100% faithful –  
*mutations* can occur when mistakes are made



# Point mutations:

## One nucleotide replaced by another

- May be SILENT/synonymous (remember codons!)

GCC (Ala) → GC**A** (Ala)

- Or may change a single amino acid (*missense mutation*)

GCC (Ala) → **C**CC (Pro)

- Or may introduce a stop codon (*nonsense mutation*)

CAA (Gln) → **T**AA (Stop)

# Other types of non-synonymous change

- Insertions

AAA ACC GAA TCA CCG GAT → AAA ACC GCA ATC ACC GGA T..

- Deletions

AAA ACC GAA TCA CCG GAT → AAA ACC GAA TCC CGG AT.

- Inversions

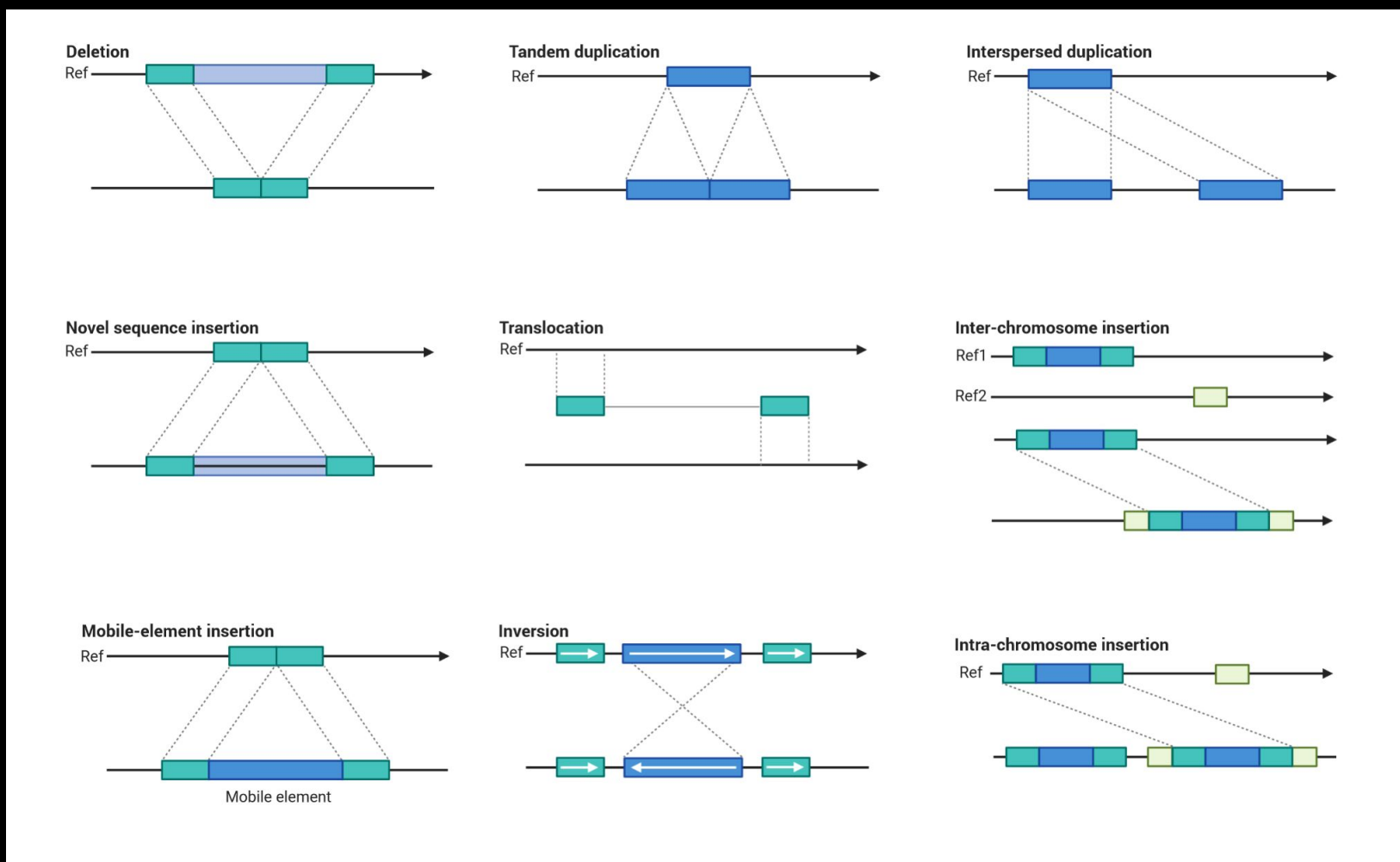
AAA ACC GAA TCA CCG GAT → AAA ACC TGA TTC CCG GAT

- Duplications (up the whole genome!)

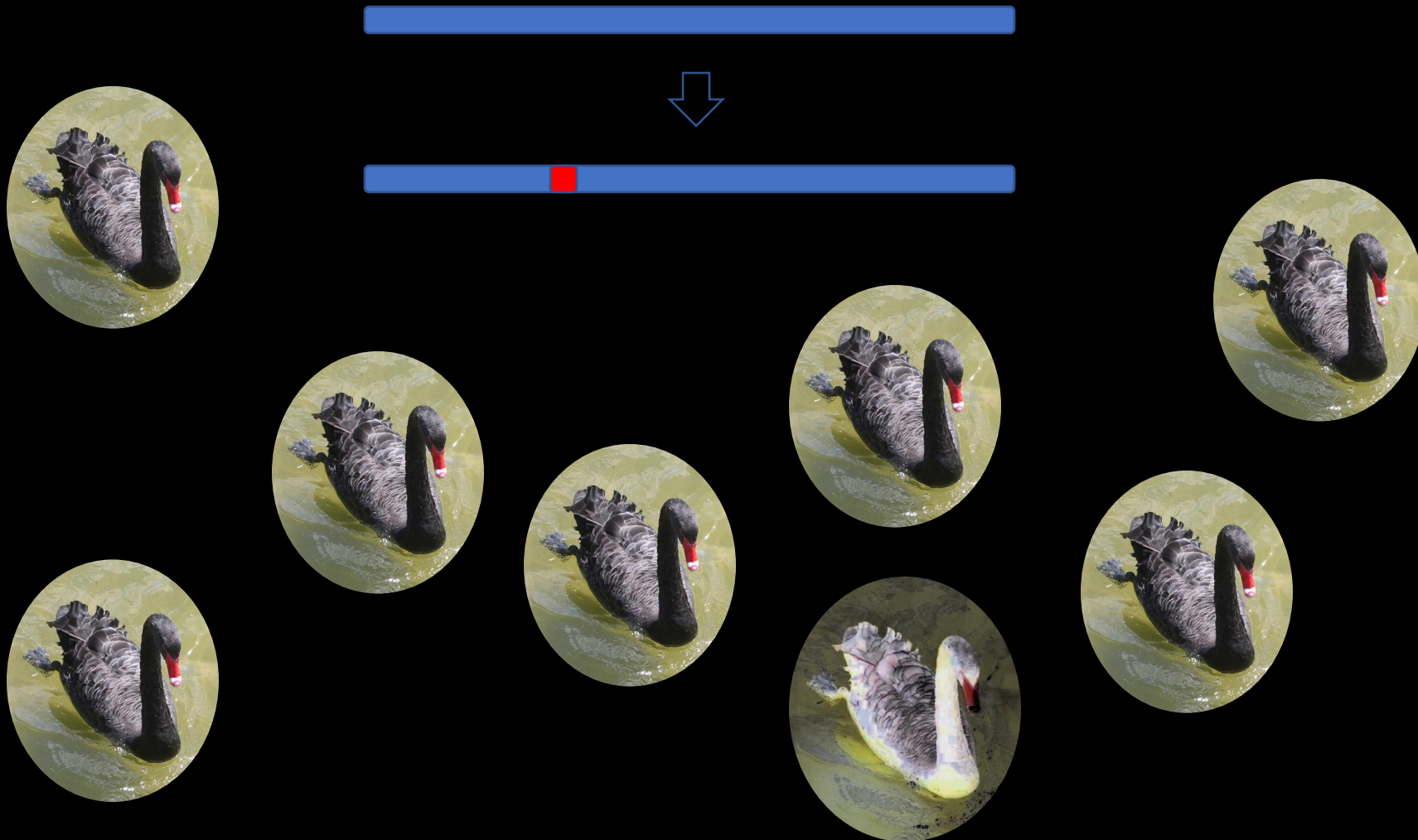
AAA ACC GAA TCA CCG GAT → AAA ACC GAA TCG AAT CAC CGG AT.

(this is why we need sequence alignment – coming in Module 2)

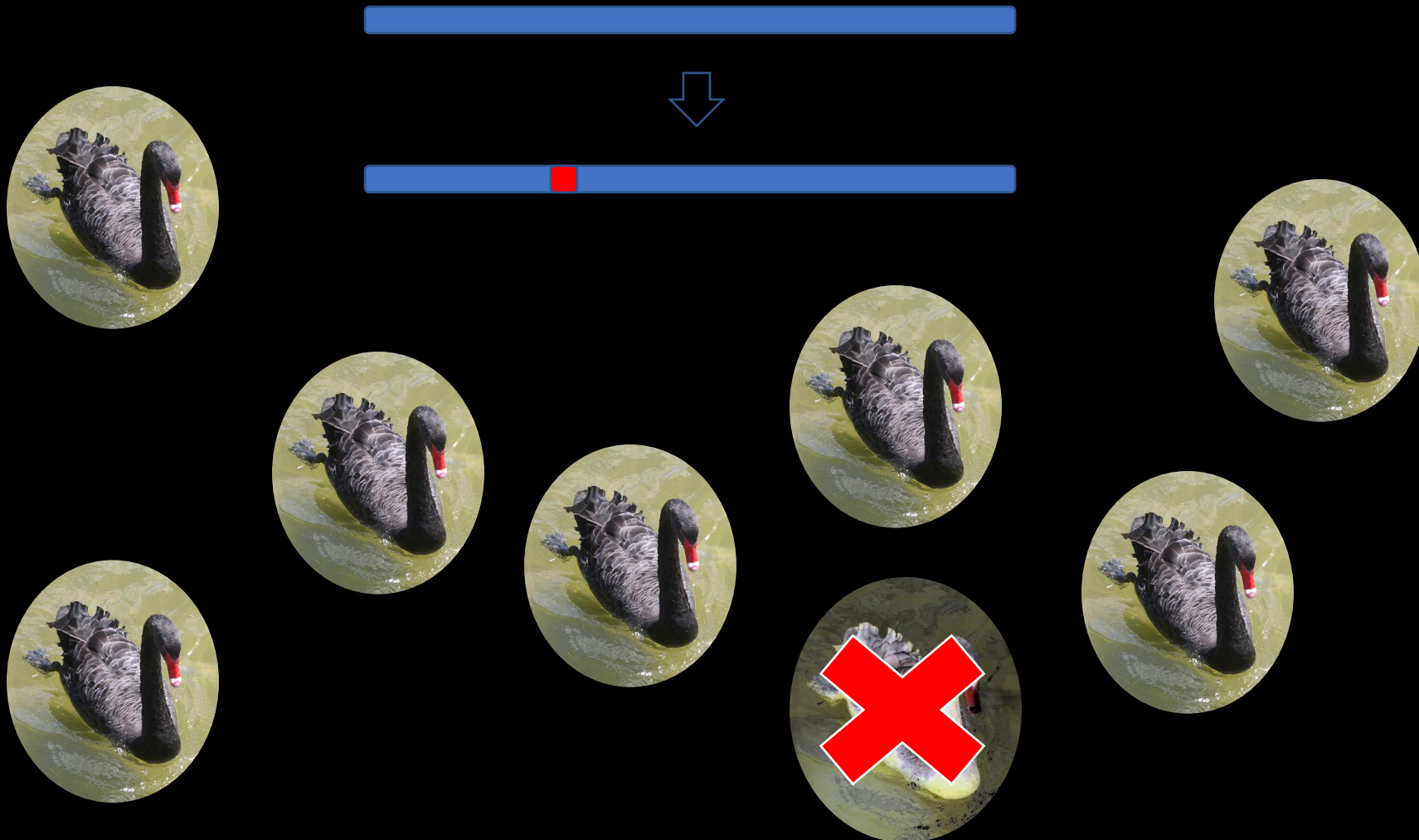
# Larger scale structural variations



# The population context of a mutation

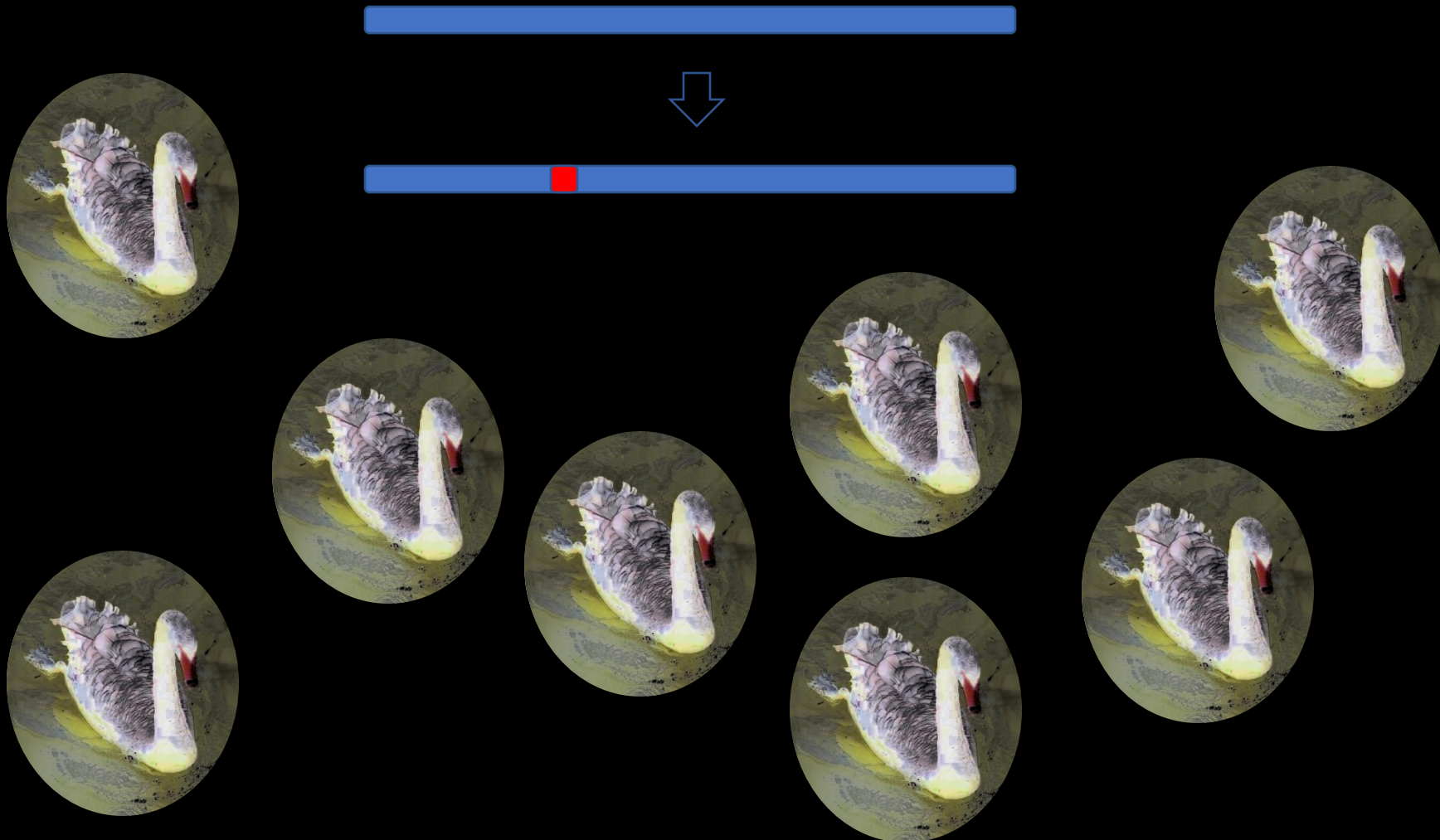


# The population context of a mutation



DELETERIOUS

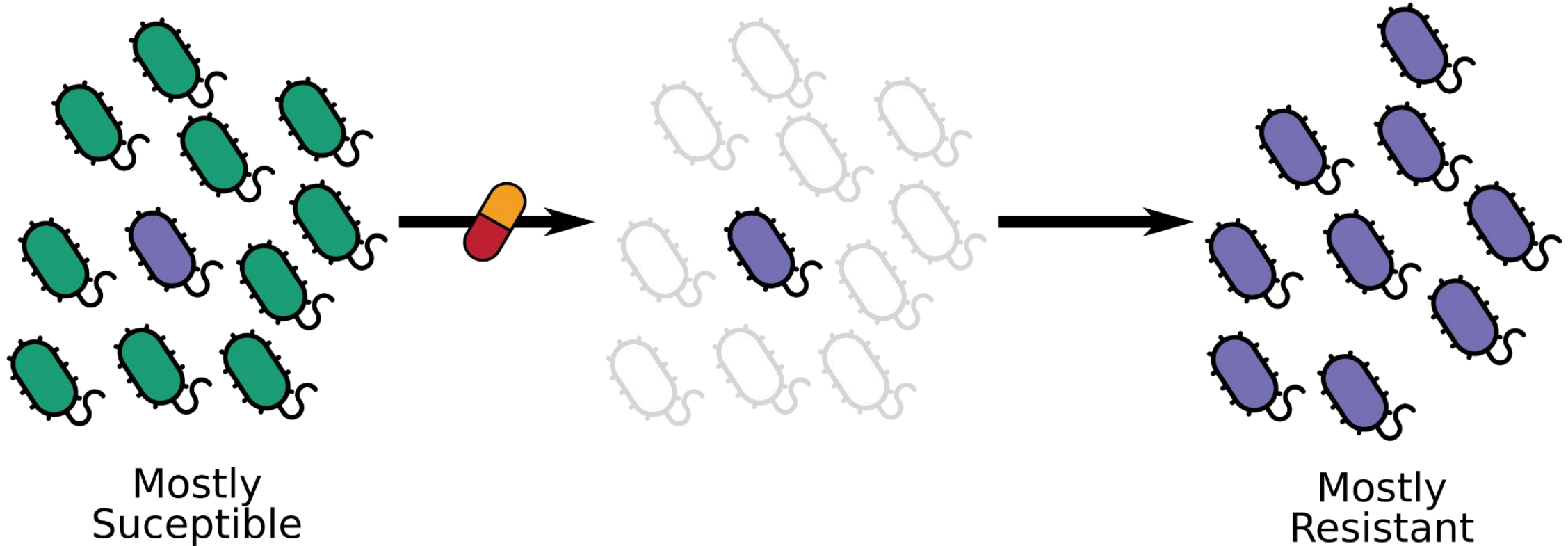
# The population context of a mutation



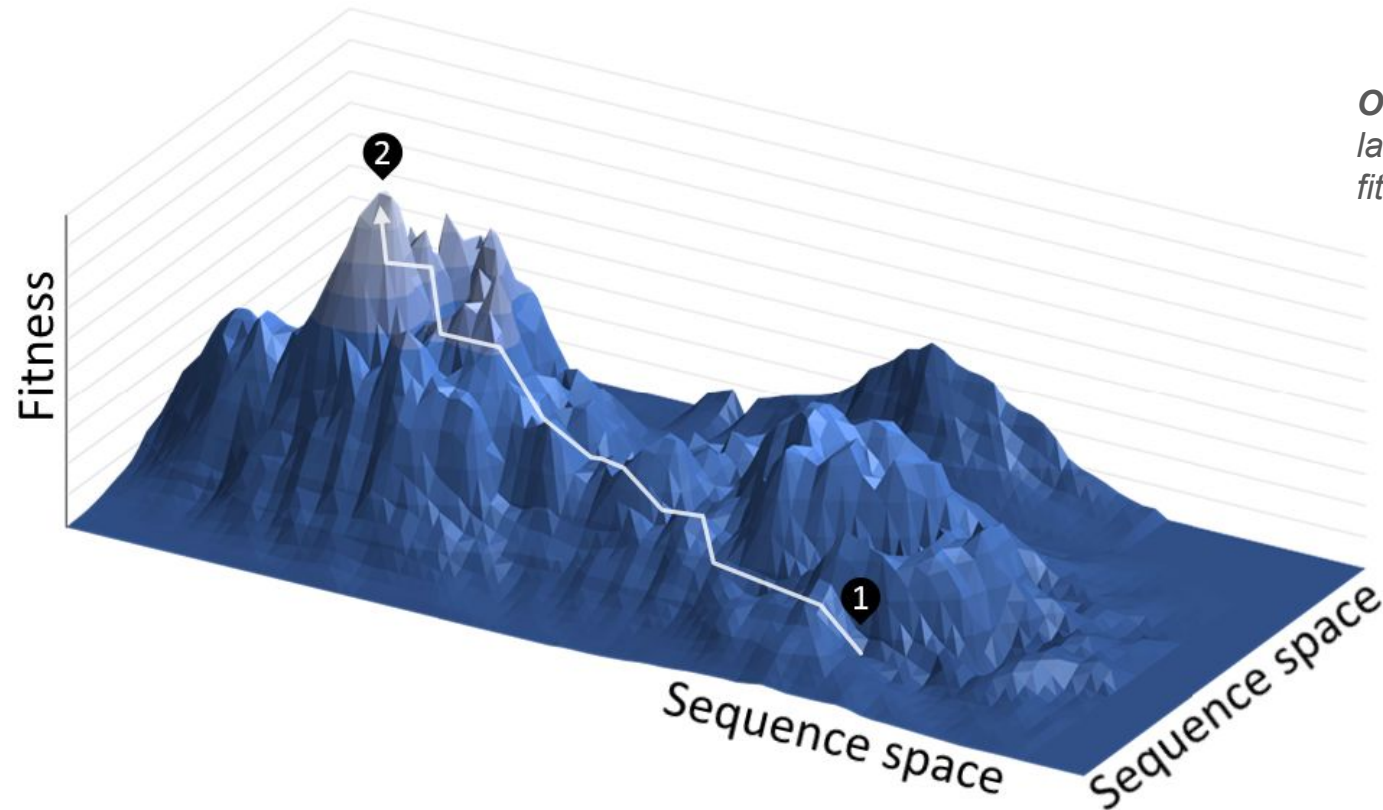
ADVANTAGEOUS



# Strong selective force rapidly changes genotype



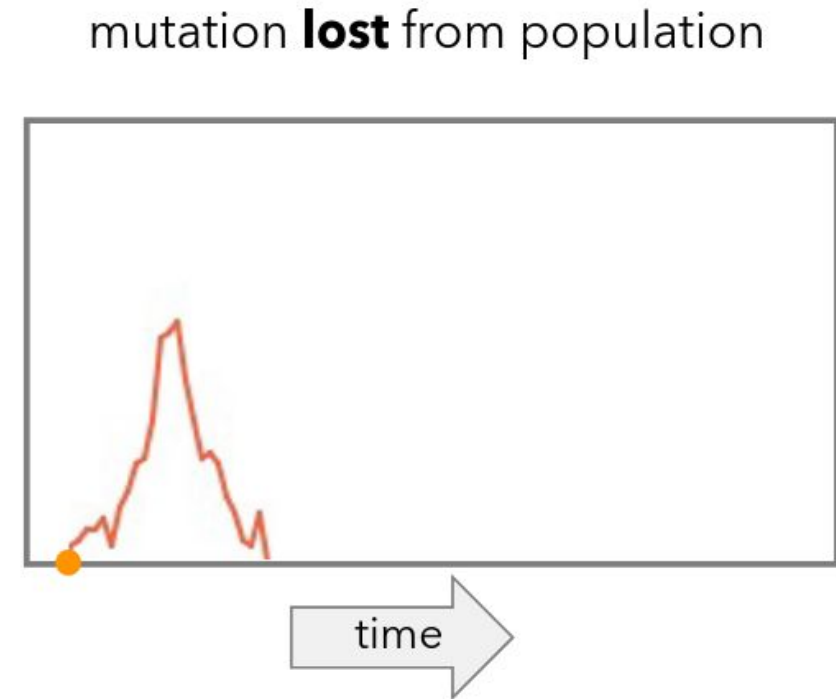
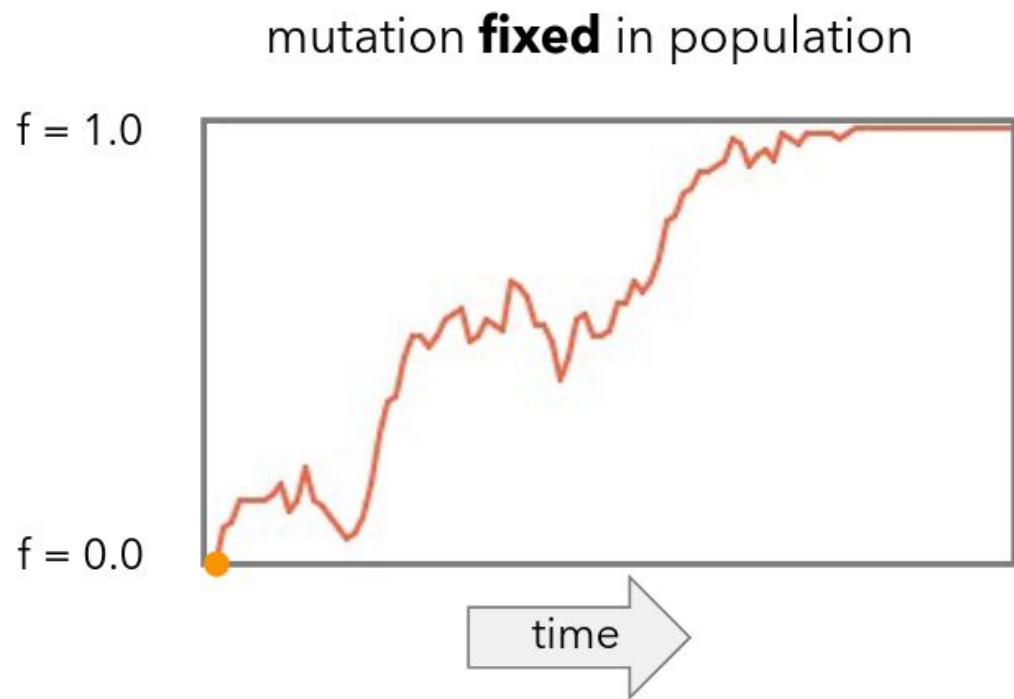
# Mutations as a random walk across a fitness landscape



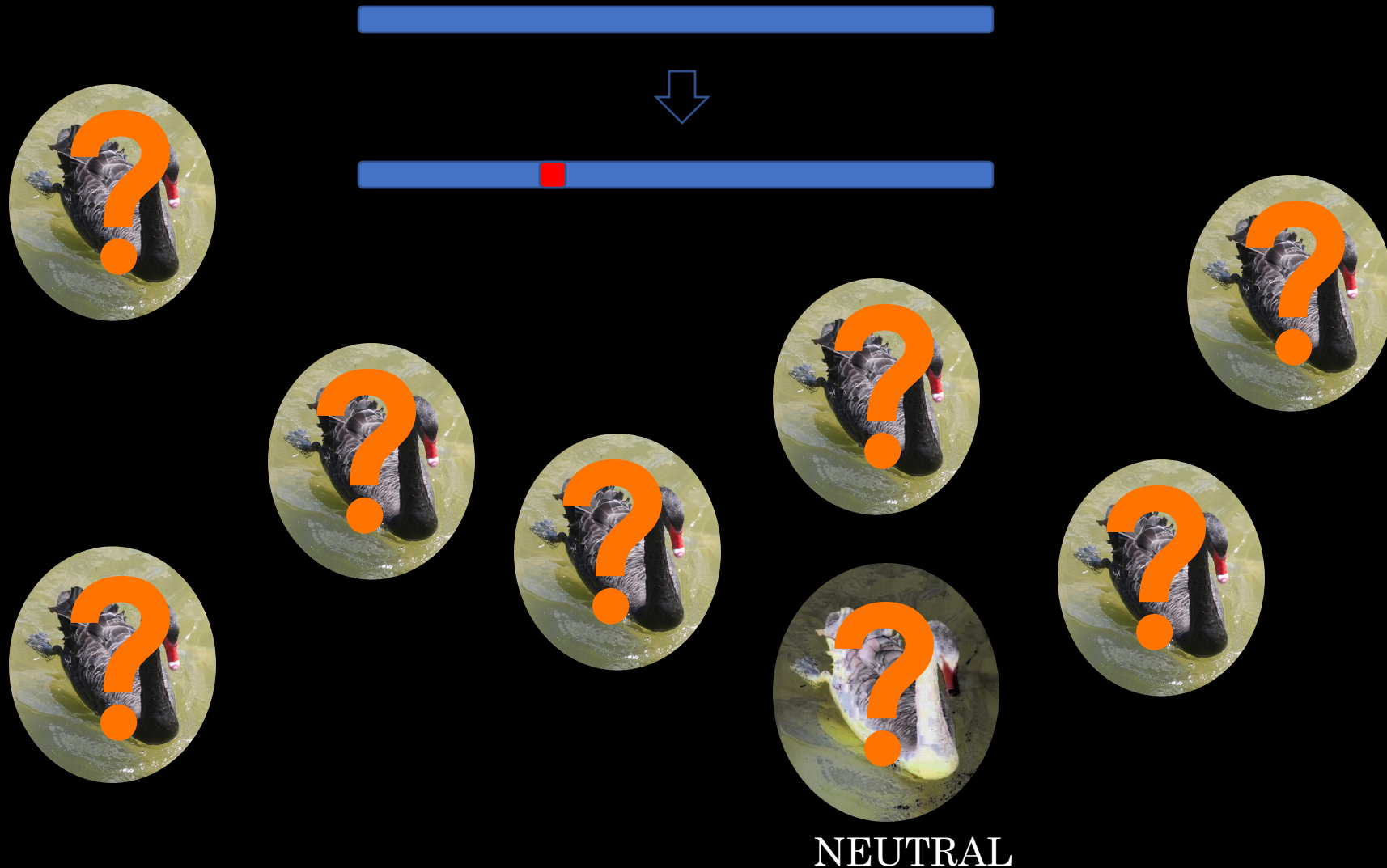
**OVERSIMPLIFICATION** - actual landscapes are dynamic/changing & fitness is hard to measure

- *Fitness = quantitative representation of individual reproductive success*
- *High Fitness = more descendants (larger proportion of circulating population)*
- *Low Fitness = fewer descendants (higher chance of dying out)*

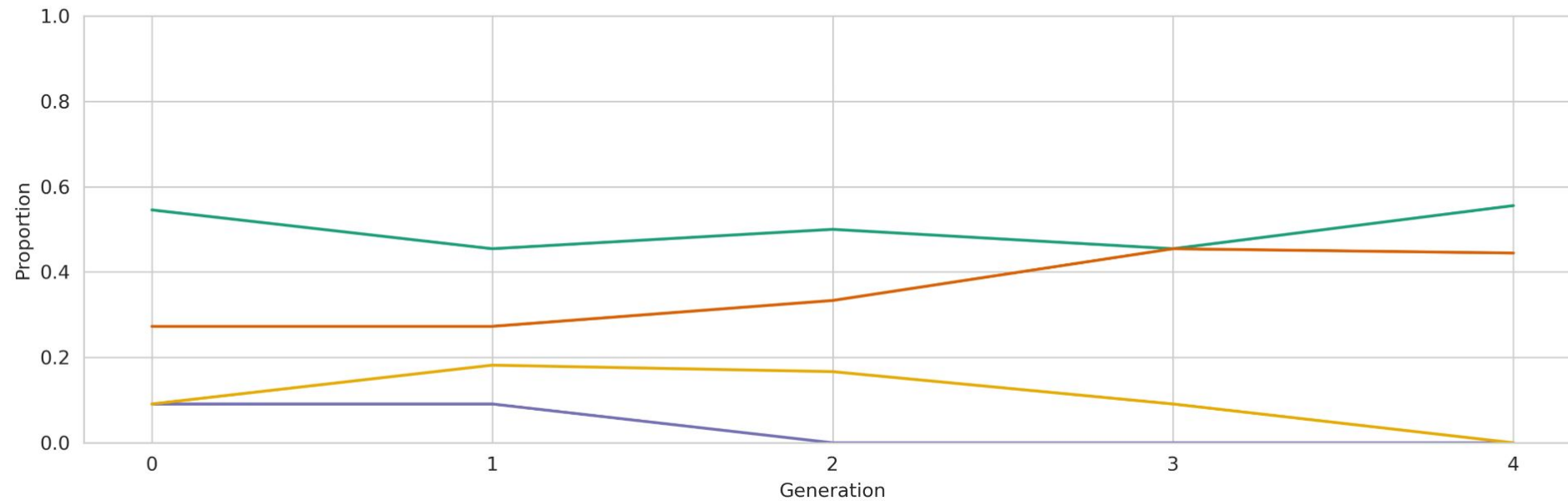
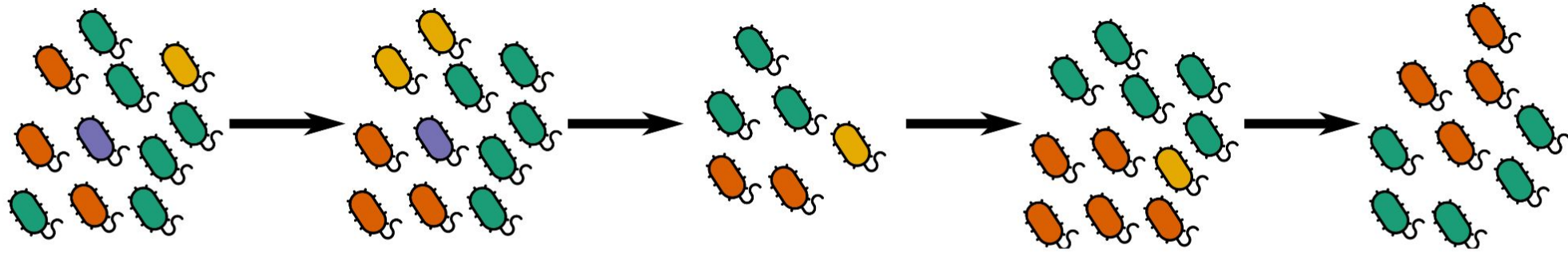
# Plotting proportion of population with a specific mutation



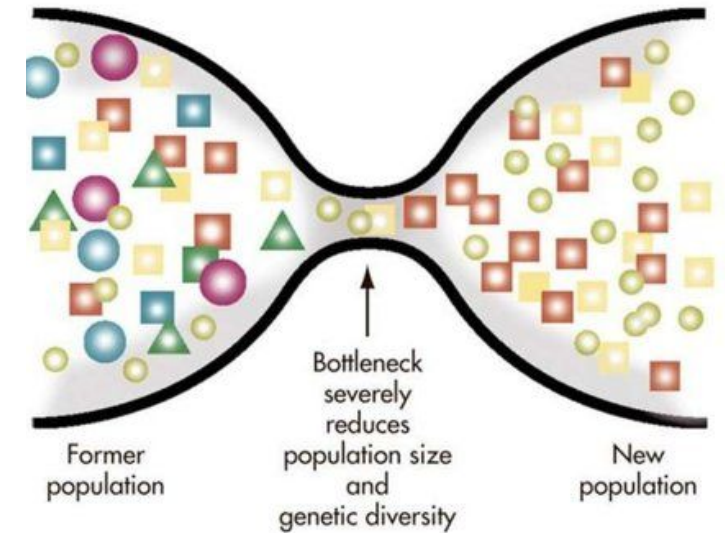
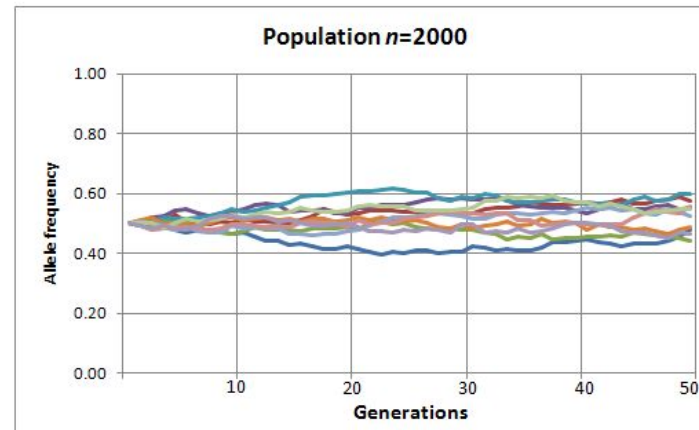
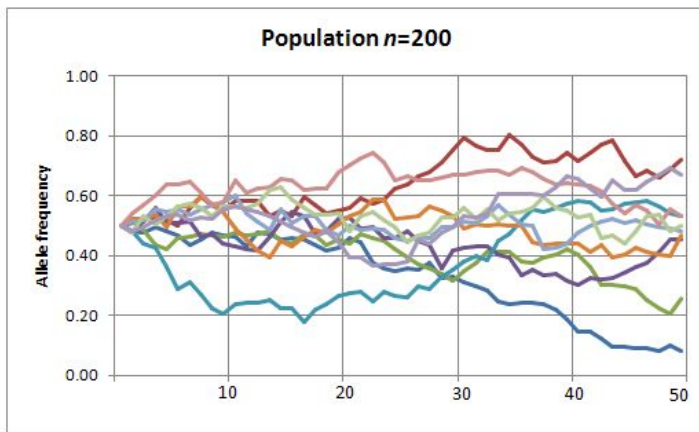
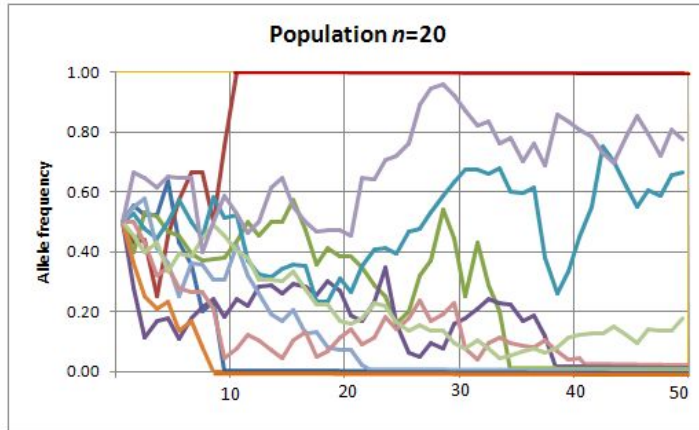
# The population context of a mutation



# Genotypes can also change due to random sampling: “drift”



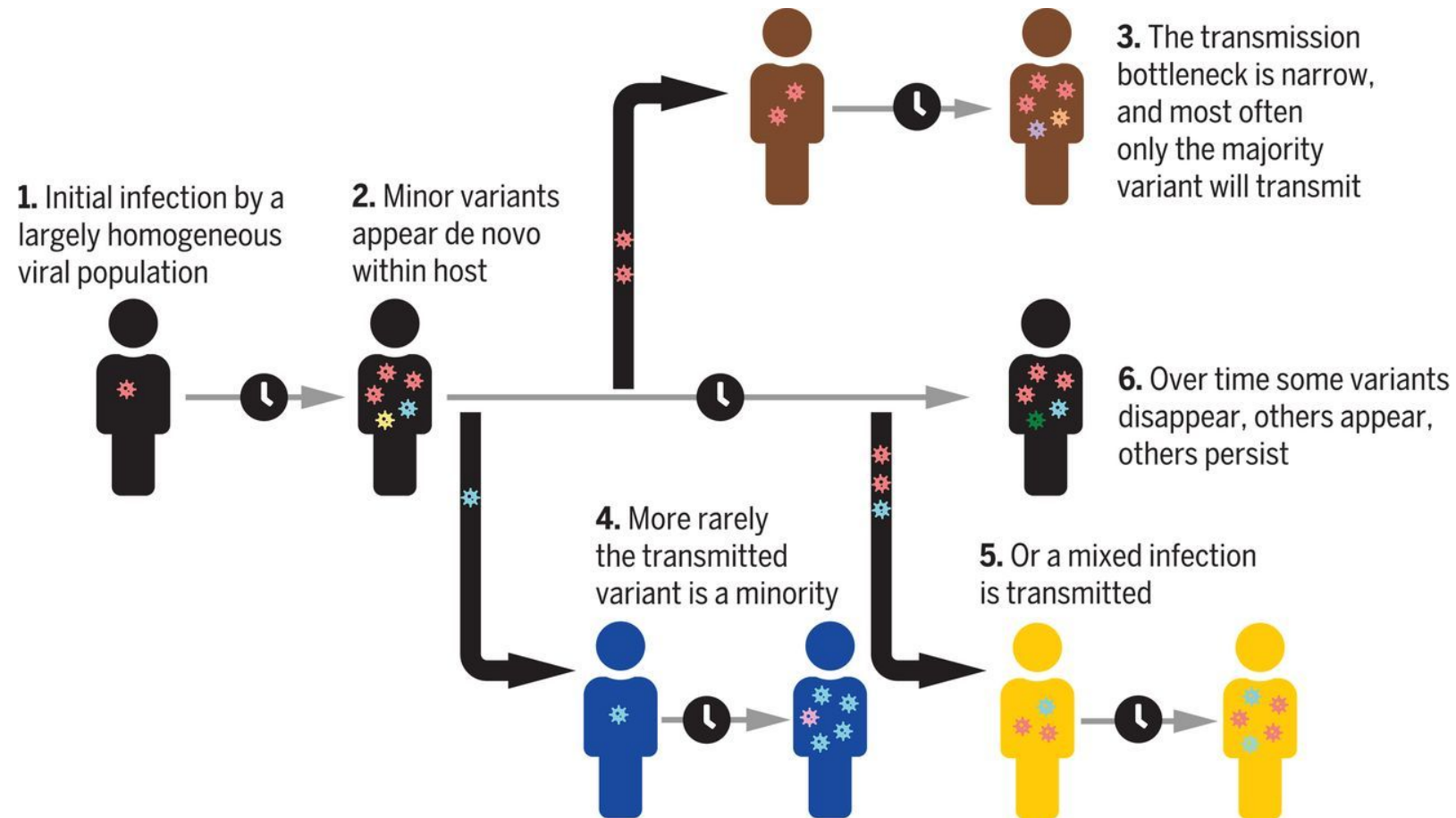
# Smaller populations are more impacted by drift



Transient reductions in population size also drive drift: “bottlenecks”



# Complicated sampling of a (within-host) population of a (between host) population



How do we estimate how much of our change is drift and how much is selection?



# dN/dS is one way to detect selection

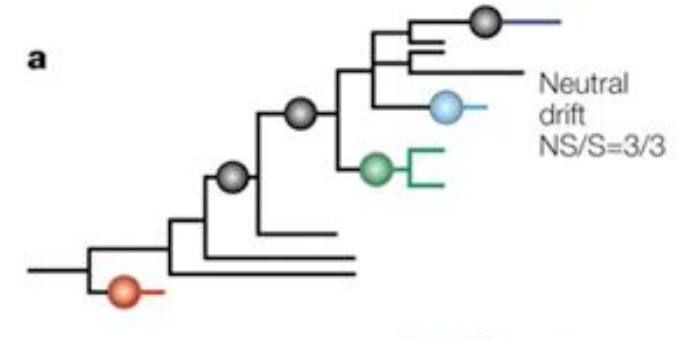
**dN** = non-synonymous mutations (normalised)

**dS** = synonymous mutations (normalised)

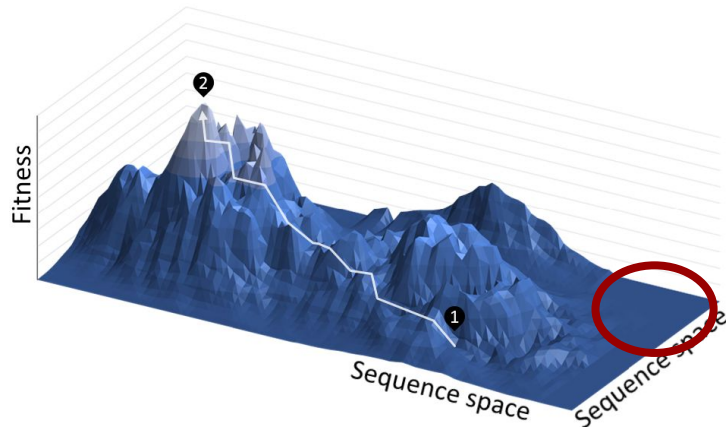
# dN/dS is one way to detect selection

**dN** = non-synonymous mutations (normalised)

**dS** = synonymous mutations (normalised)



**dN/dS ~ 1** : drift/neutral selection



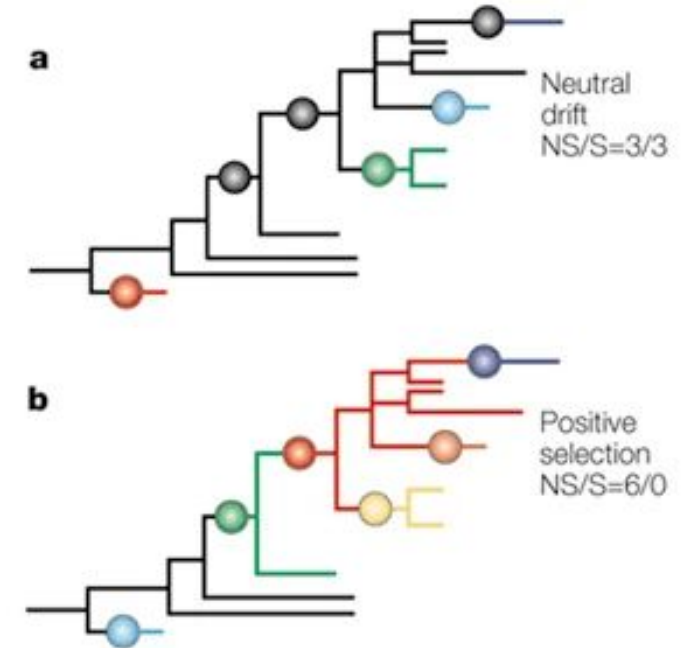
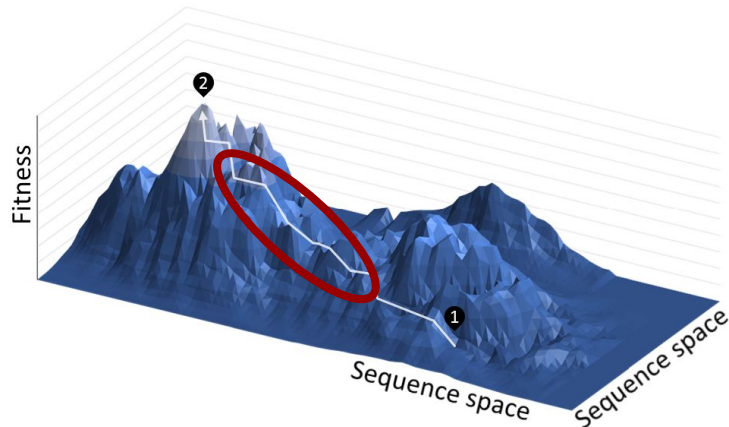
# dN/dS is one way to detect selection

**dN** = non-synonymous mutations (normalised)

**dS** = synonymous mutations (normalised)

**dN/dS > 1** : adaptive/positive selection

**dN/dS ~ 1** : drift/neutral selection



# dN/dS is one way to detect selection

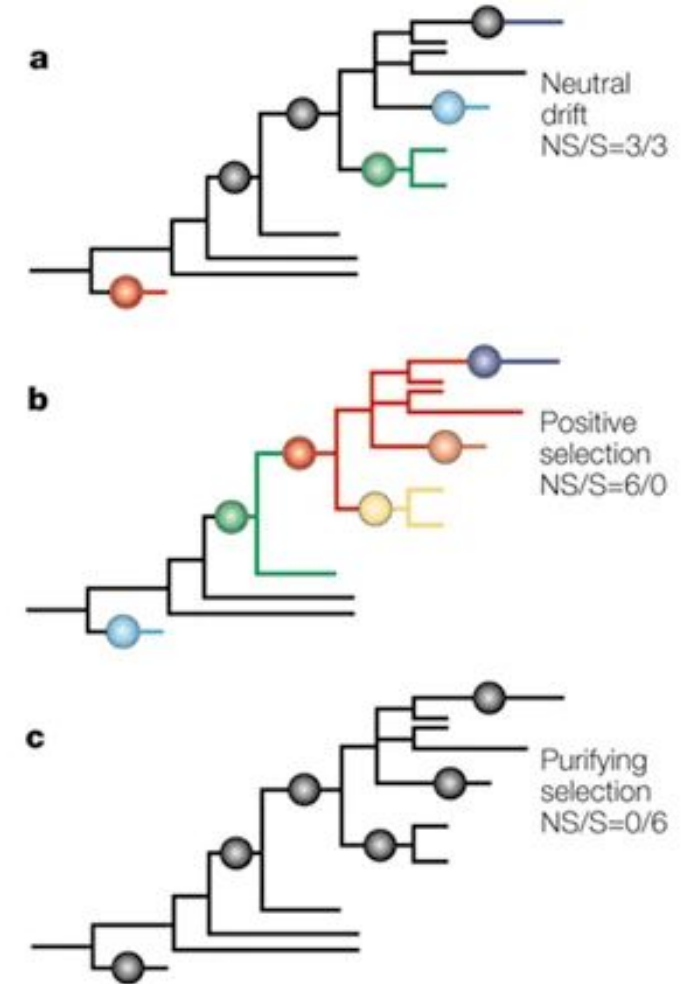
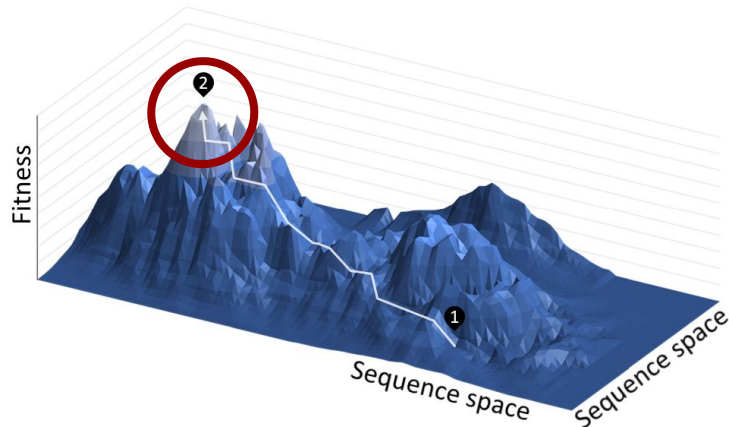
**dN** = non-synonymous mutations (normalised)

**dS** = synonymous mutations (normalised)

$dN/dS > 1$  : adaptive/positive selection

$dN/dS \sim 1$  : drift/neutral selection

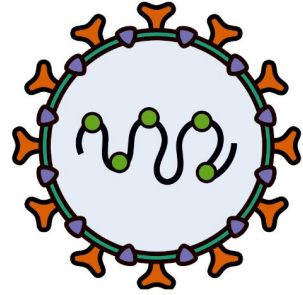
$dN/dS < 1$  : purifying/negative selection



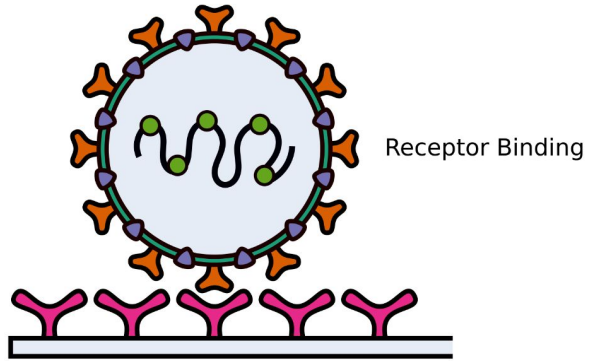
Nature Reviews | **Genetics**

Using these approaches to investigate if shortening treatment causes problems of antiviral resistance

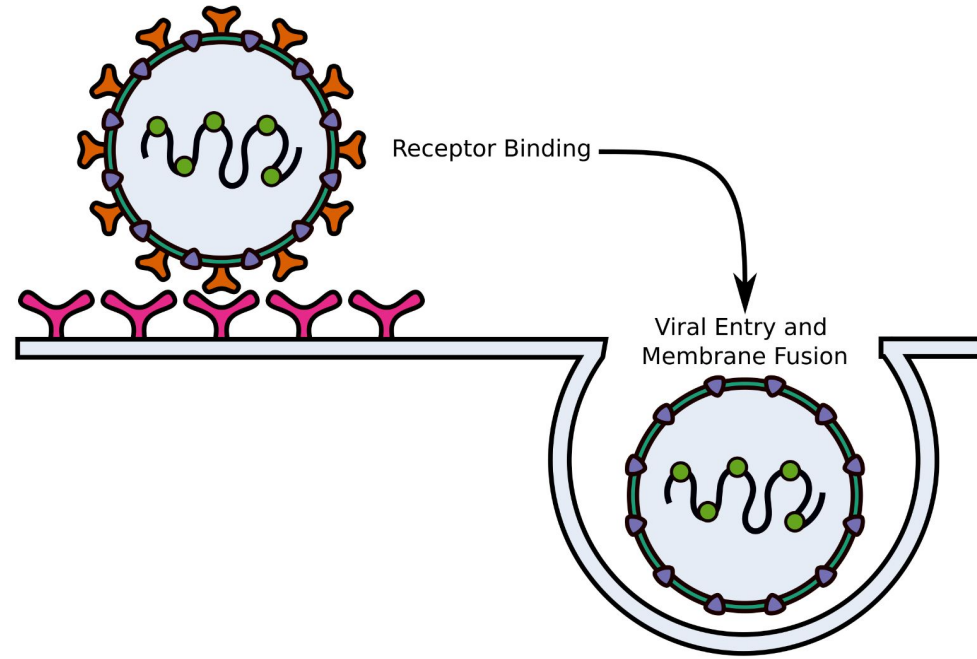
Remdesivir is an important antiviral



# Remdesivir is an important antiviral

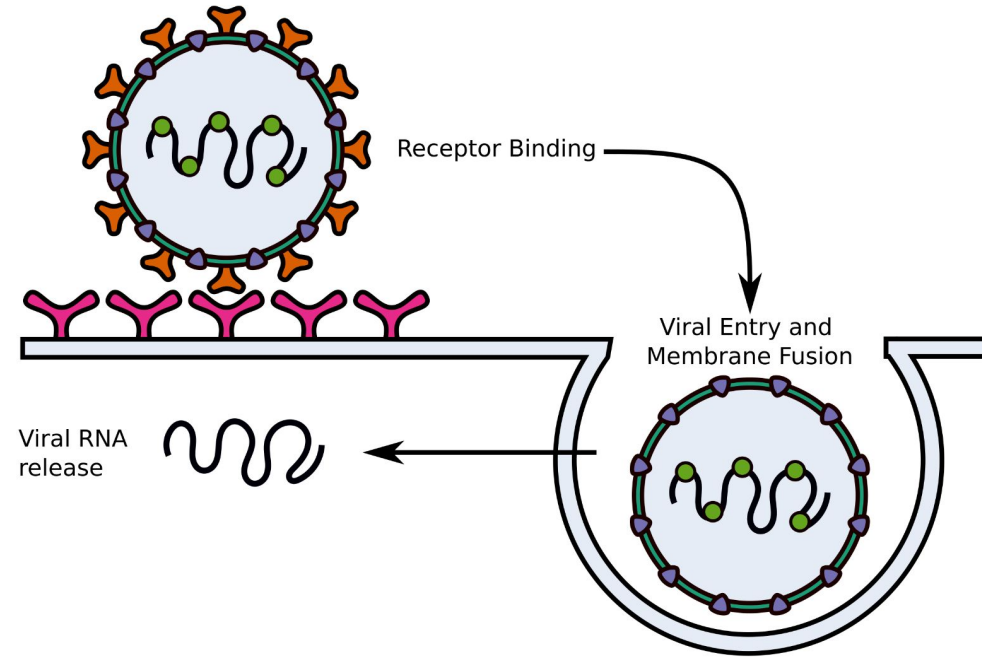


# Remdesivir is an important antiviral

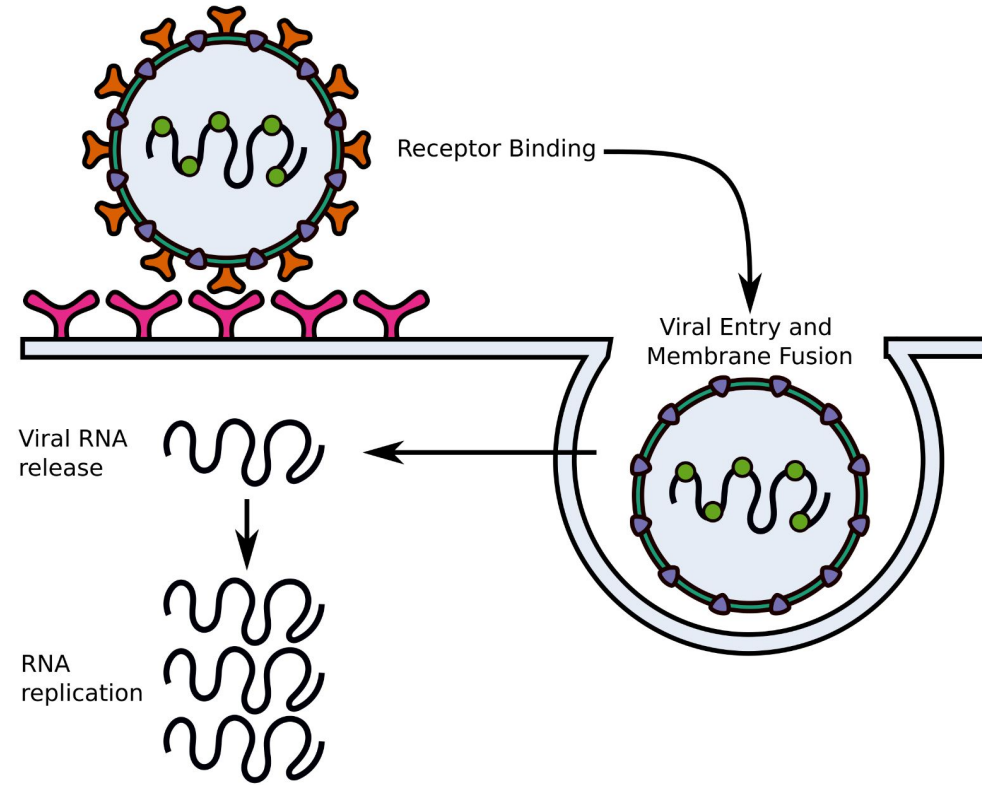




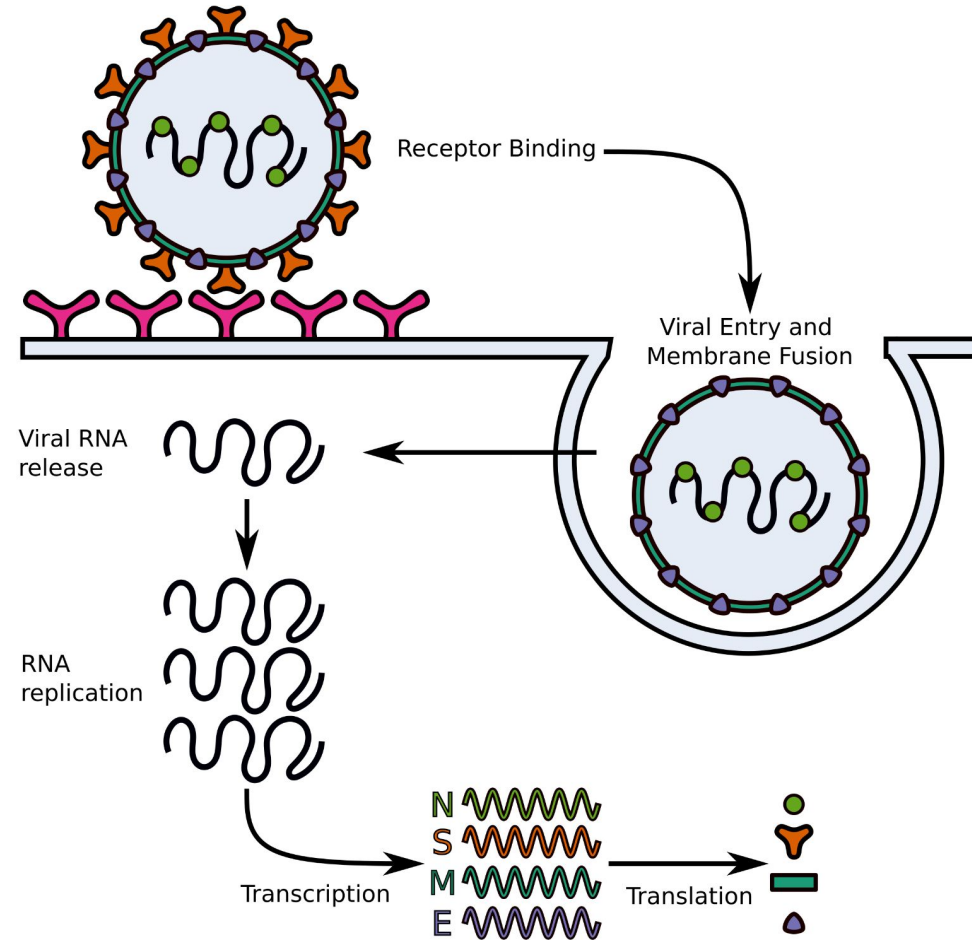
# Remdesivir is an important antiviral



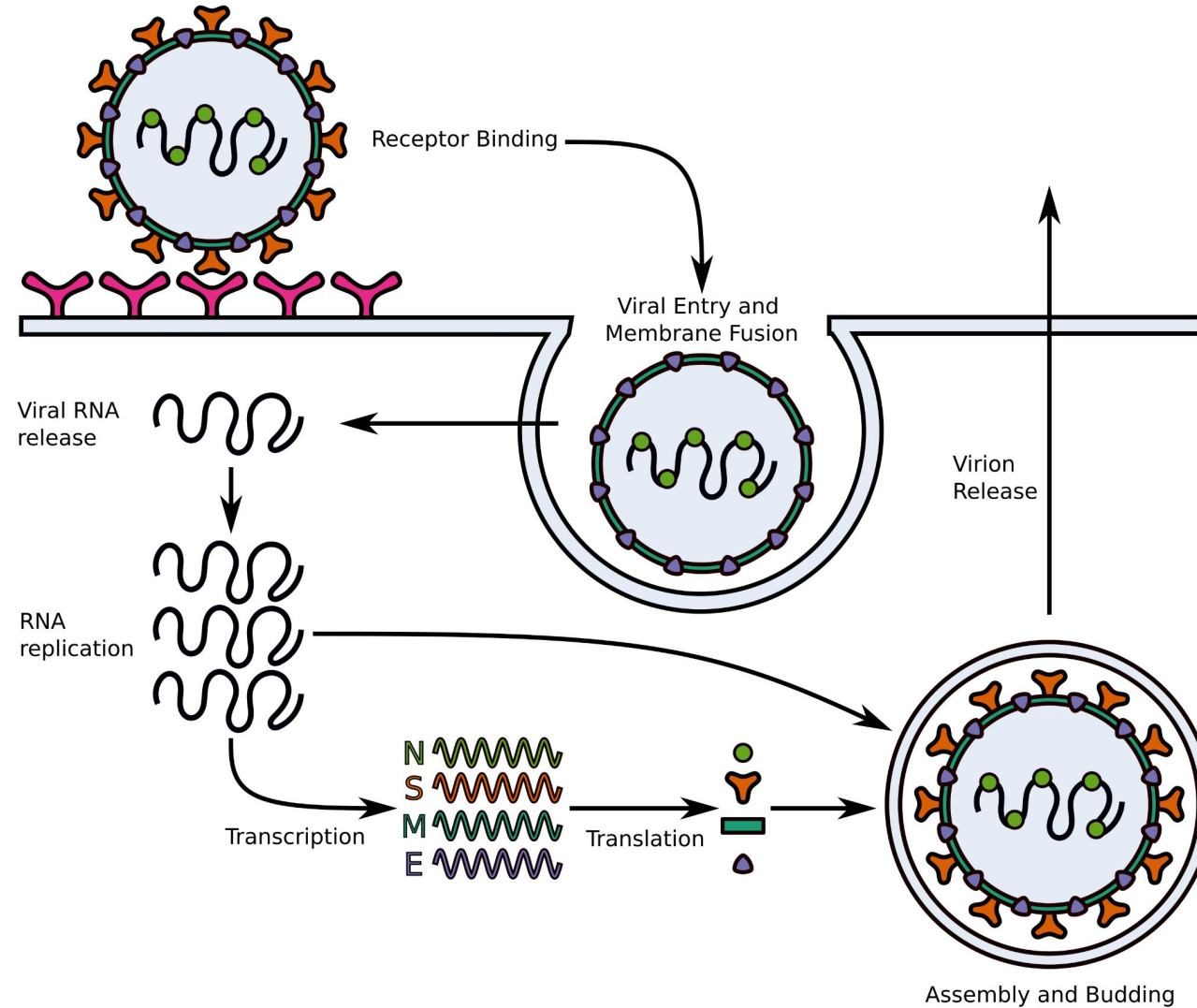
# Remdesivir is an important antiviral



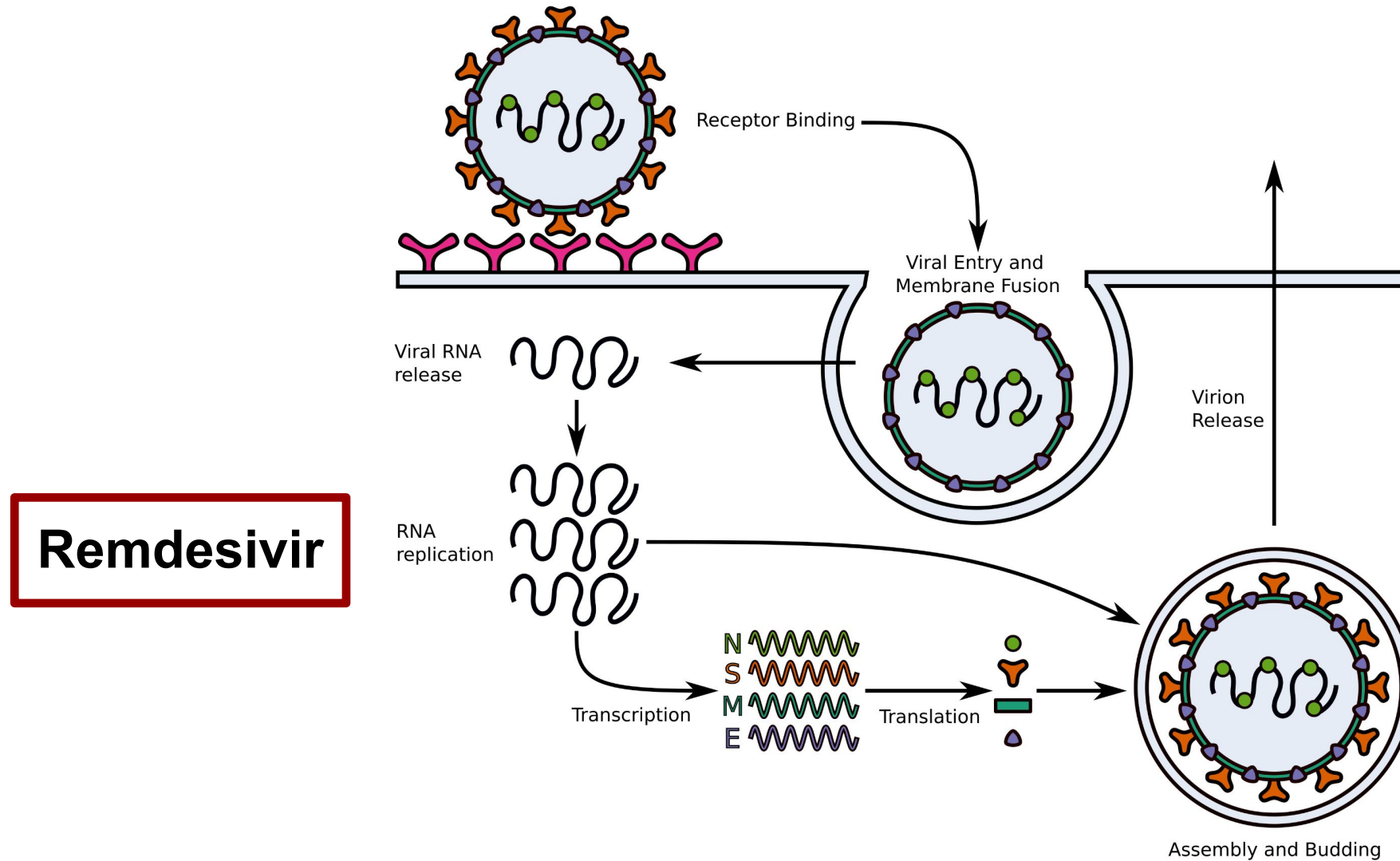
# Remdesivir is an important antiviral



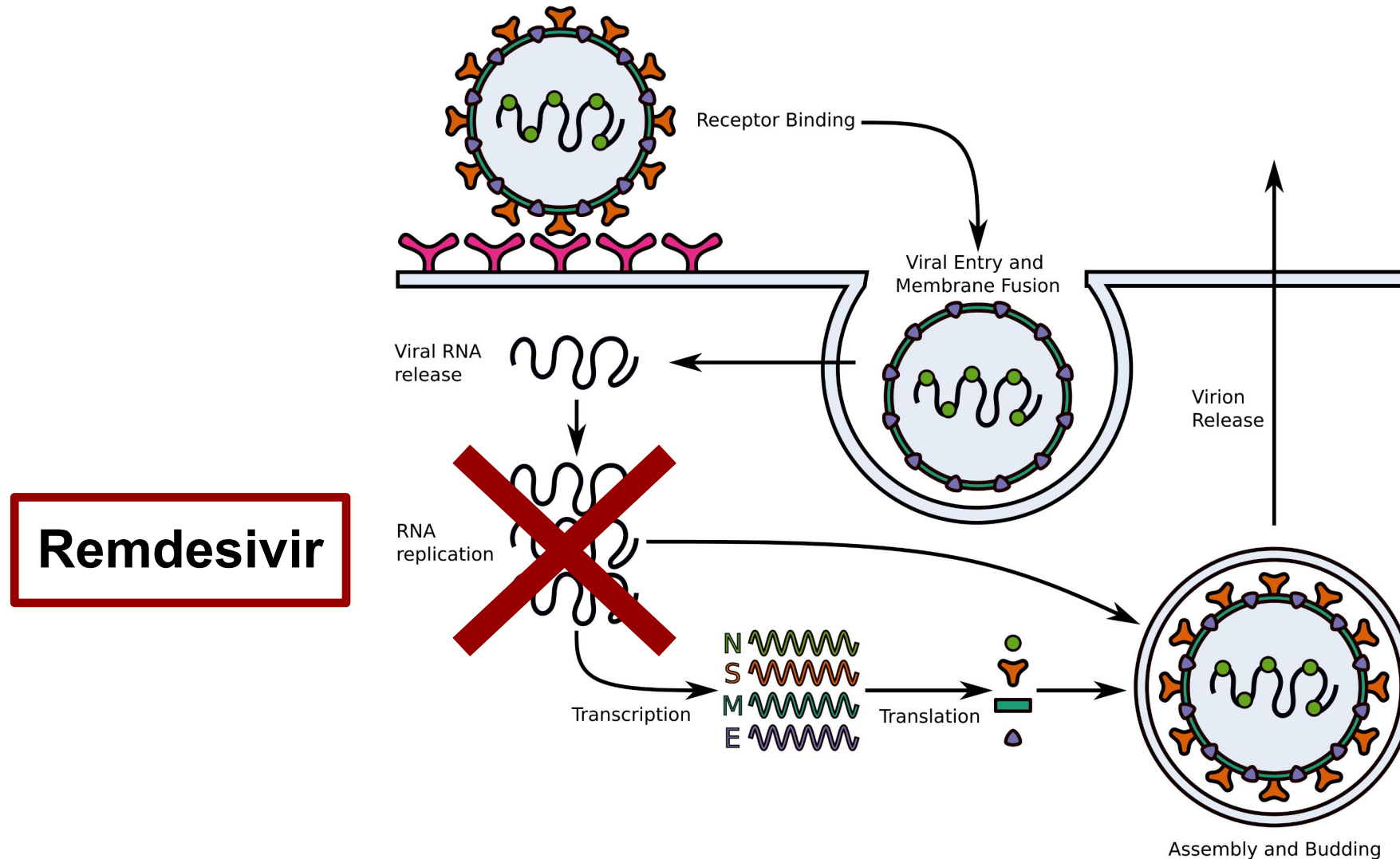
# Remdesivir is an important antiviral



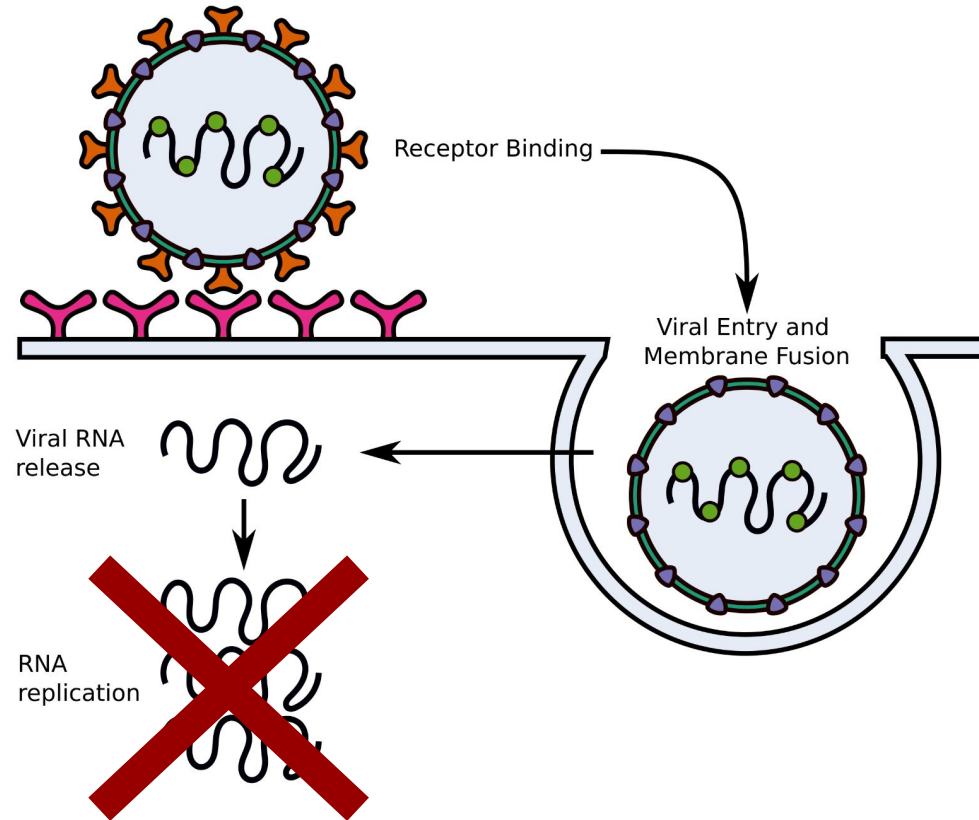
# Remdesivir is an important antiviral



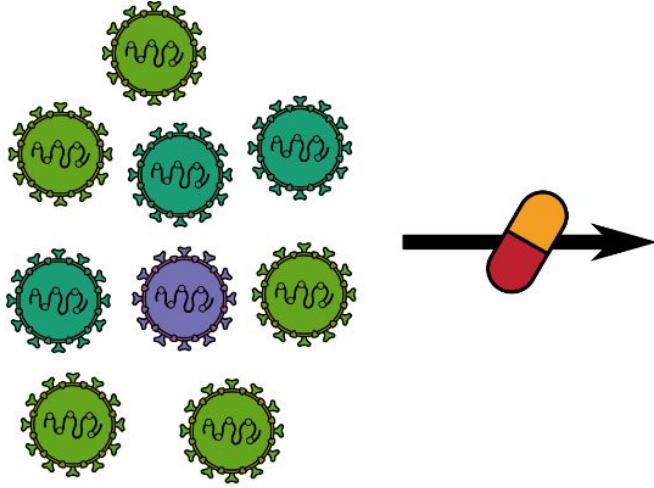
# Remdesivir is an important antiviral



# Remdesivir is an important antiviral



# Shortened treatment can promote resistance



No resistance



Moderate resistance



High resistance

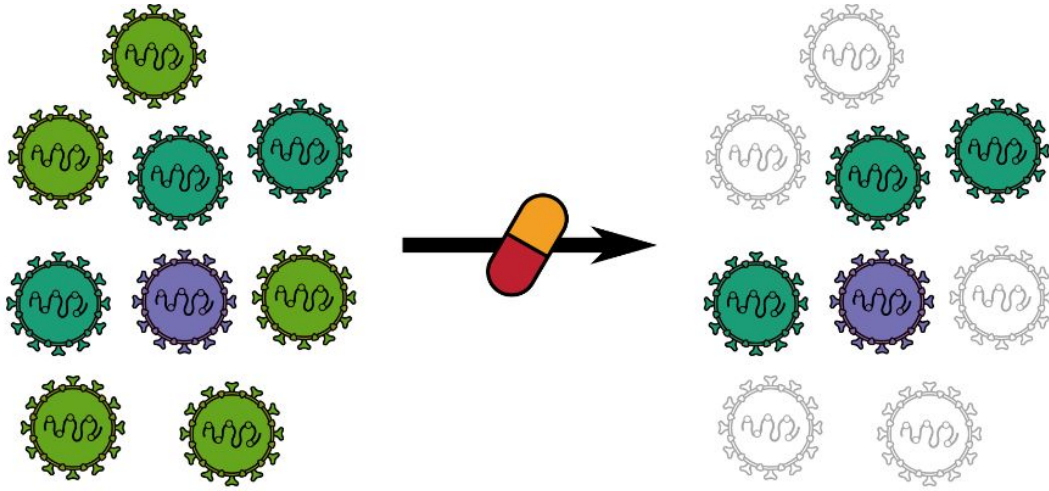


Dead

Caveat: lots of other dynamics/forces contribute to emergence of resistance (e.g., relative fitness, infection dynamics, use-patterns/contact demographics)

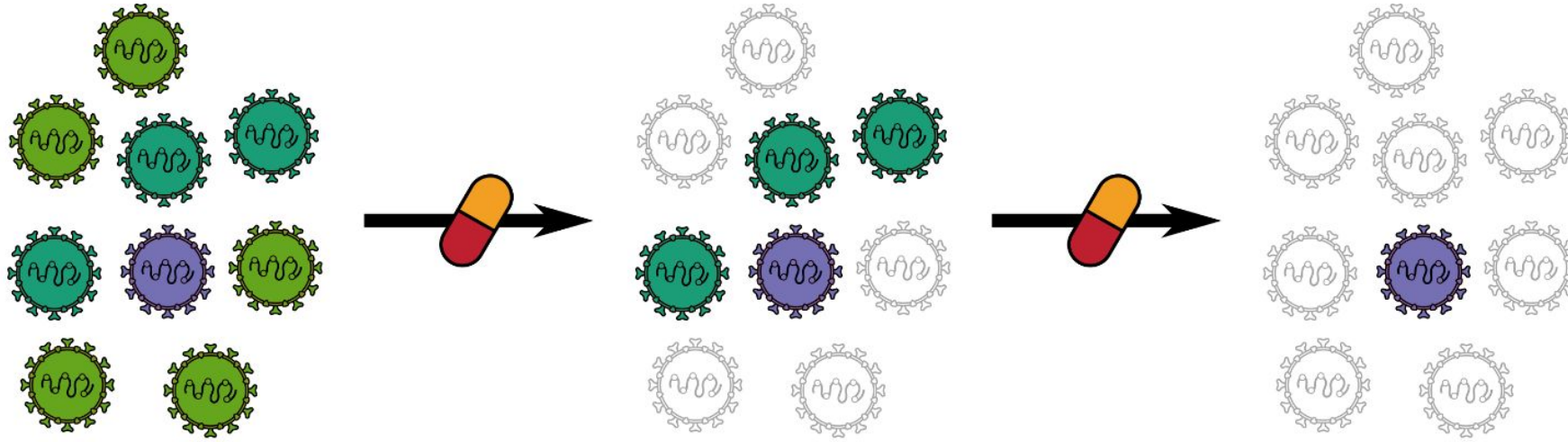


# Shortened treatment can promote resistance



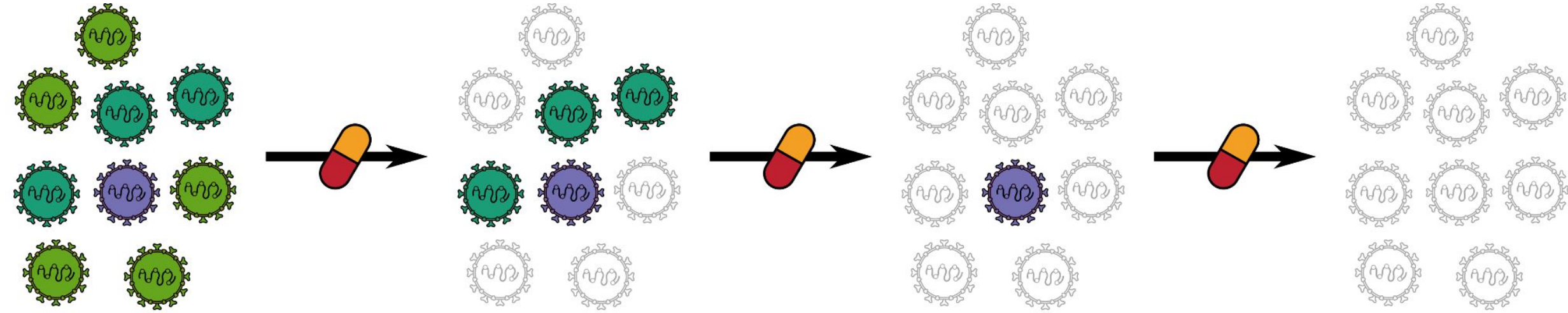
Caveat: lots of other dynamics/forces contribute to emergence of resistance (e.g., relative fitness, infection dynamics, use-patterns/contact demographics)

# Shortened treatment can promote resistance



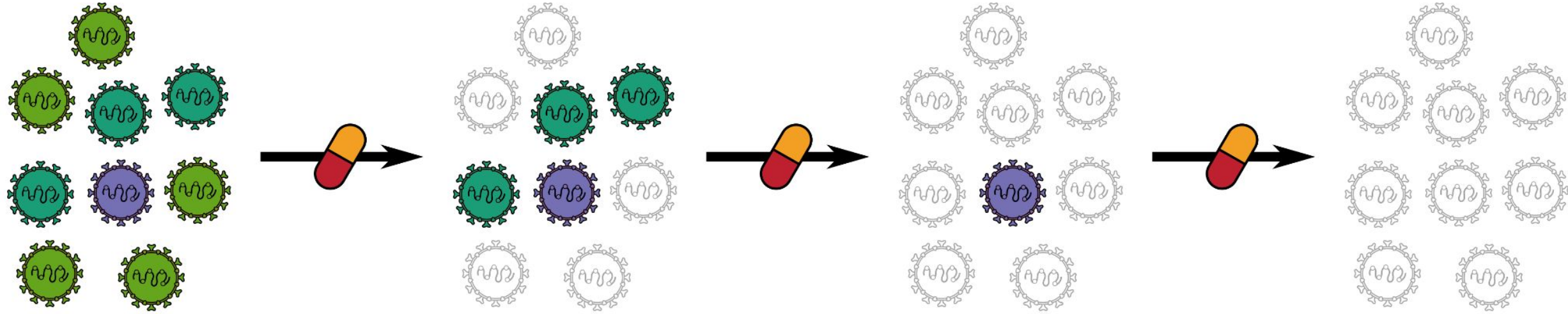
Caveat: lots of other dynamics/forces contribute to emergence of resistance (e.g., relative fitness, infection dynamics, use-patterns/contact demographics)

# Shortened treatment can promote resistance



Caveat: lots of other dynamics/forces contribute to emergence of resistance (e.g., relative fitness, infection dynamics, use-patterns/contact demographics)

# Shortened treatment can promote resistance



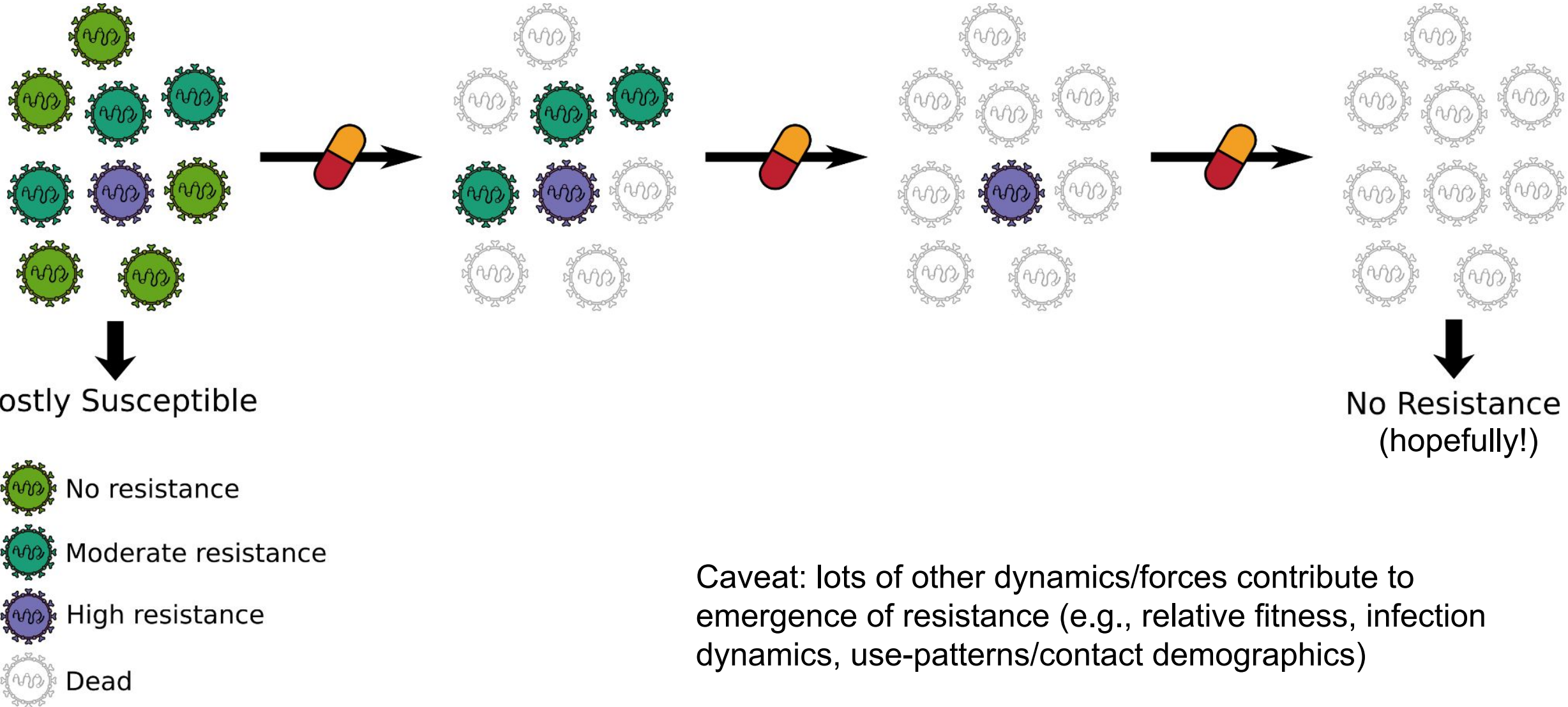
↓  
Mostly Susceptible



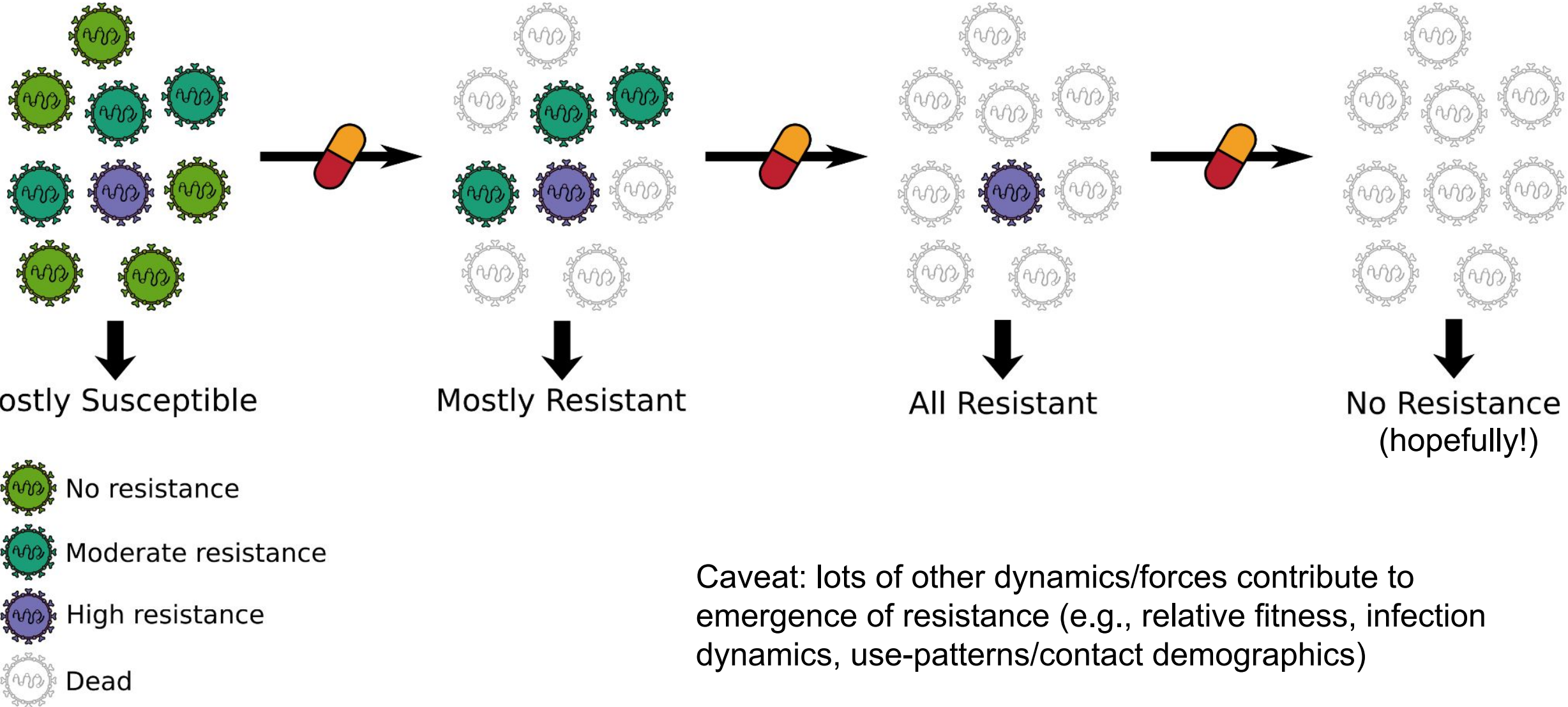
Caveat: lots of other dynamics/forces contribute to emergence of resistance (e.g., relative fitness, infection dynamics, use-patterns/contact demographics)



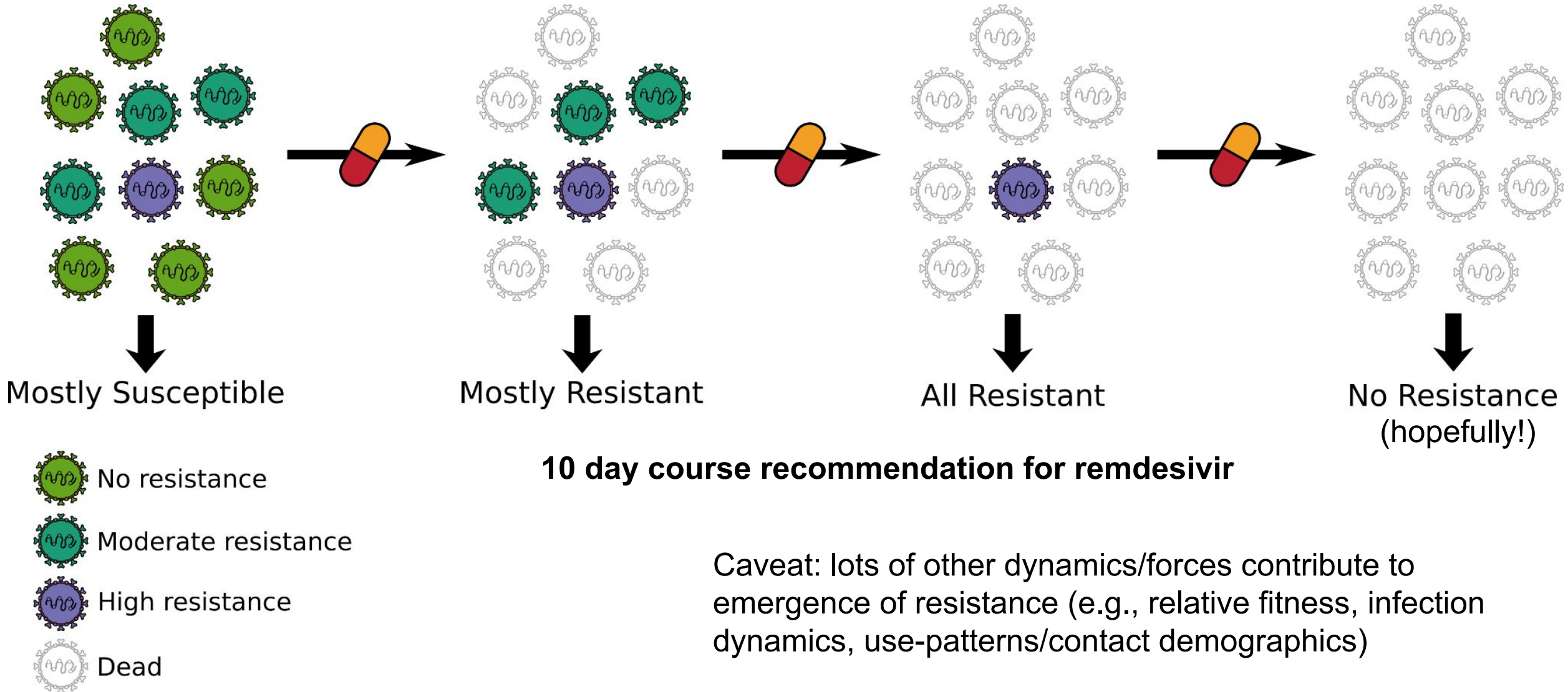
# Shortened treatment can promote resistance



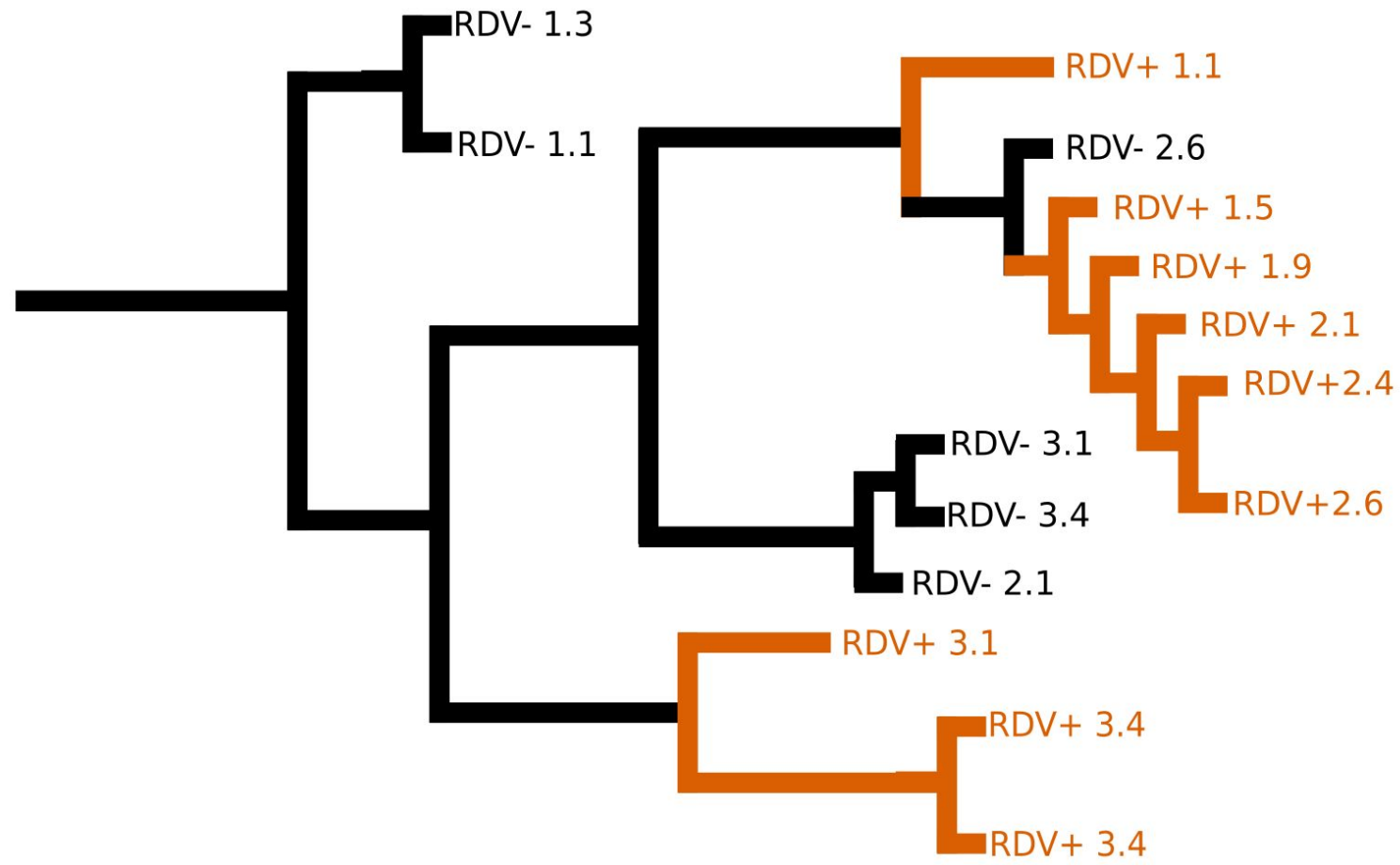
# Shortened treatment can promote resistance



# Shortened treatment can promote resistance

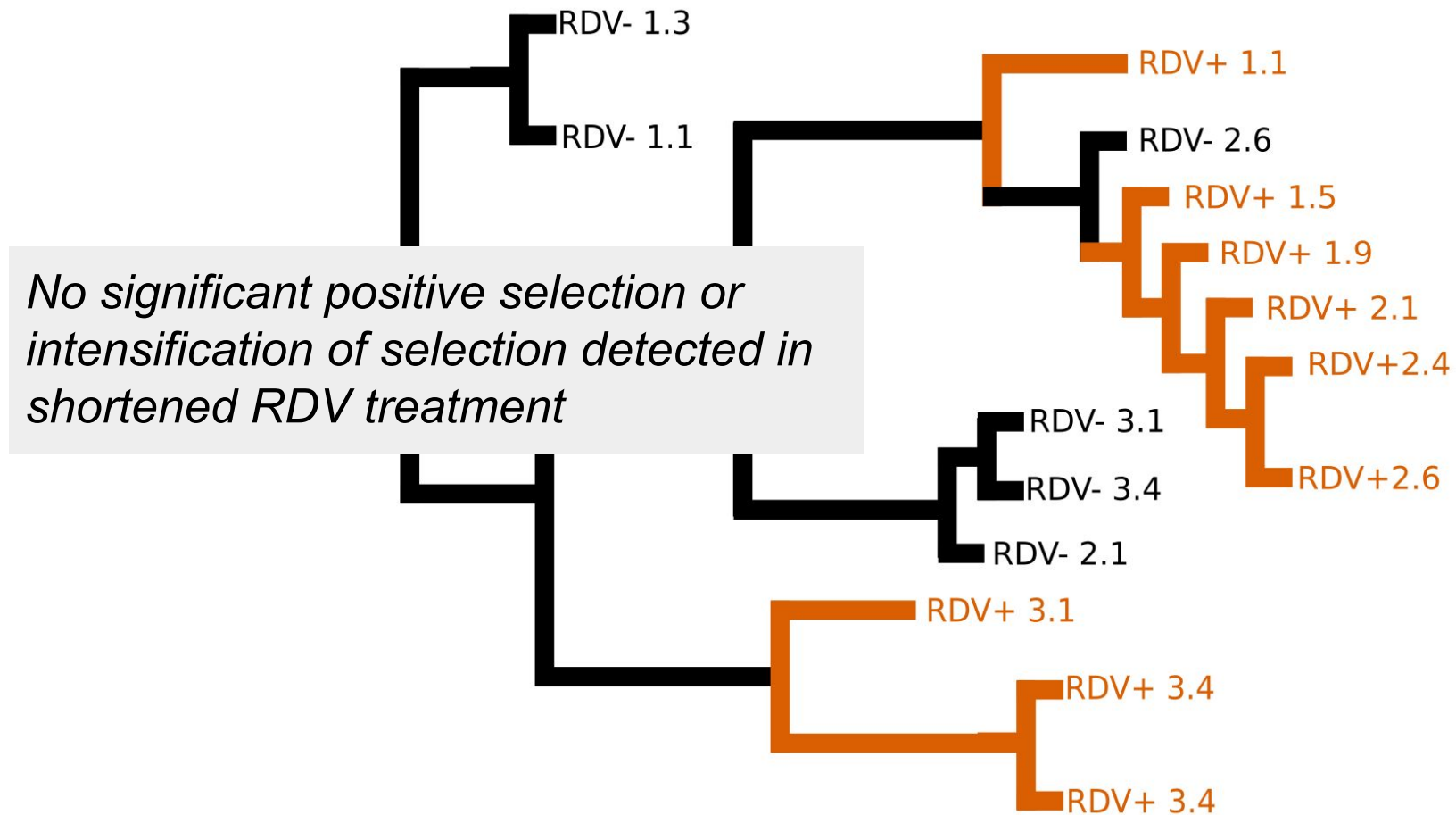


# dN/dS ratio comparison in short vs long RDV treatment

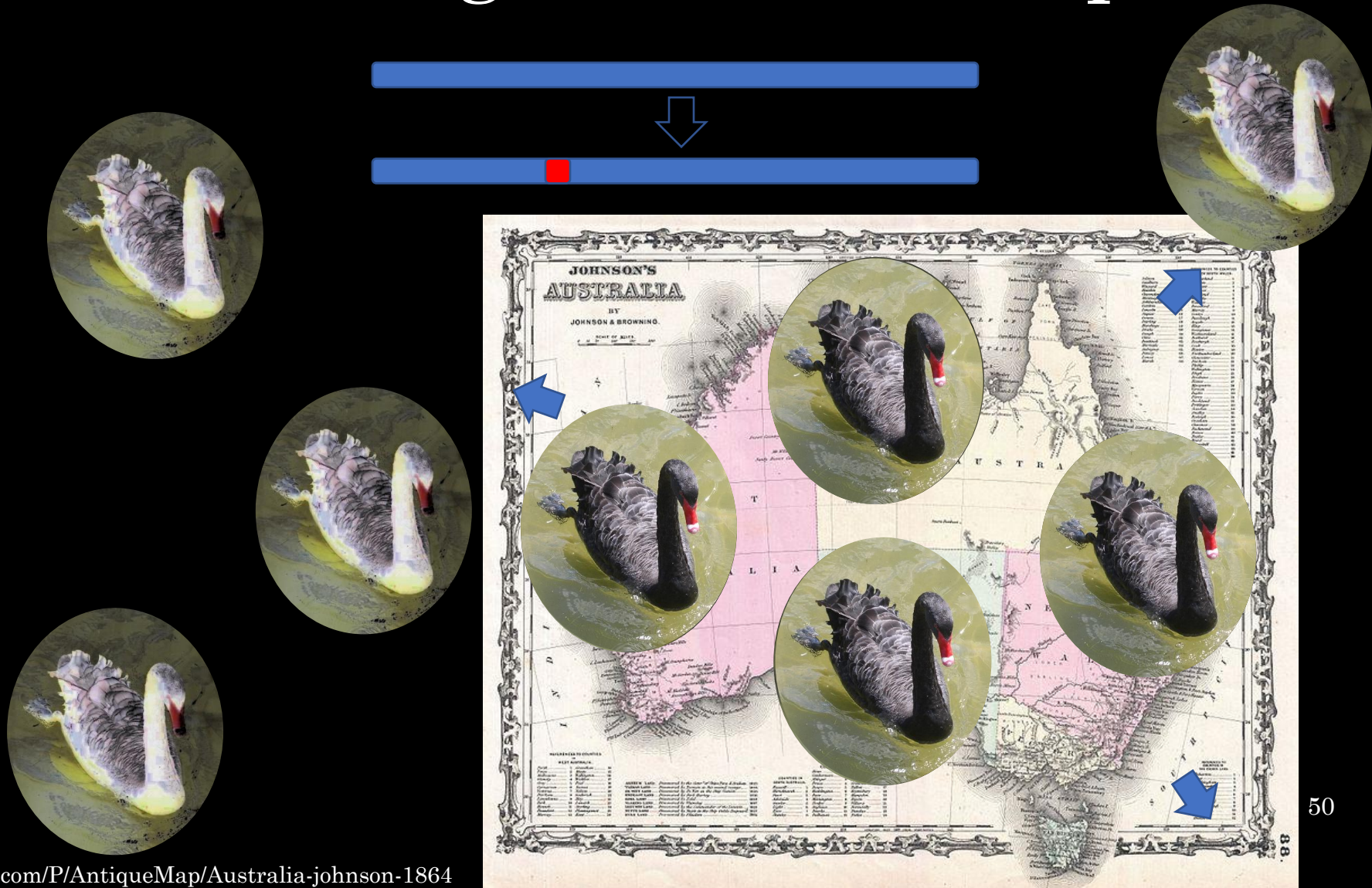




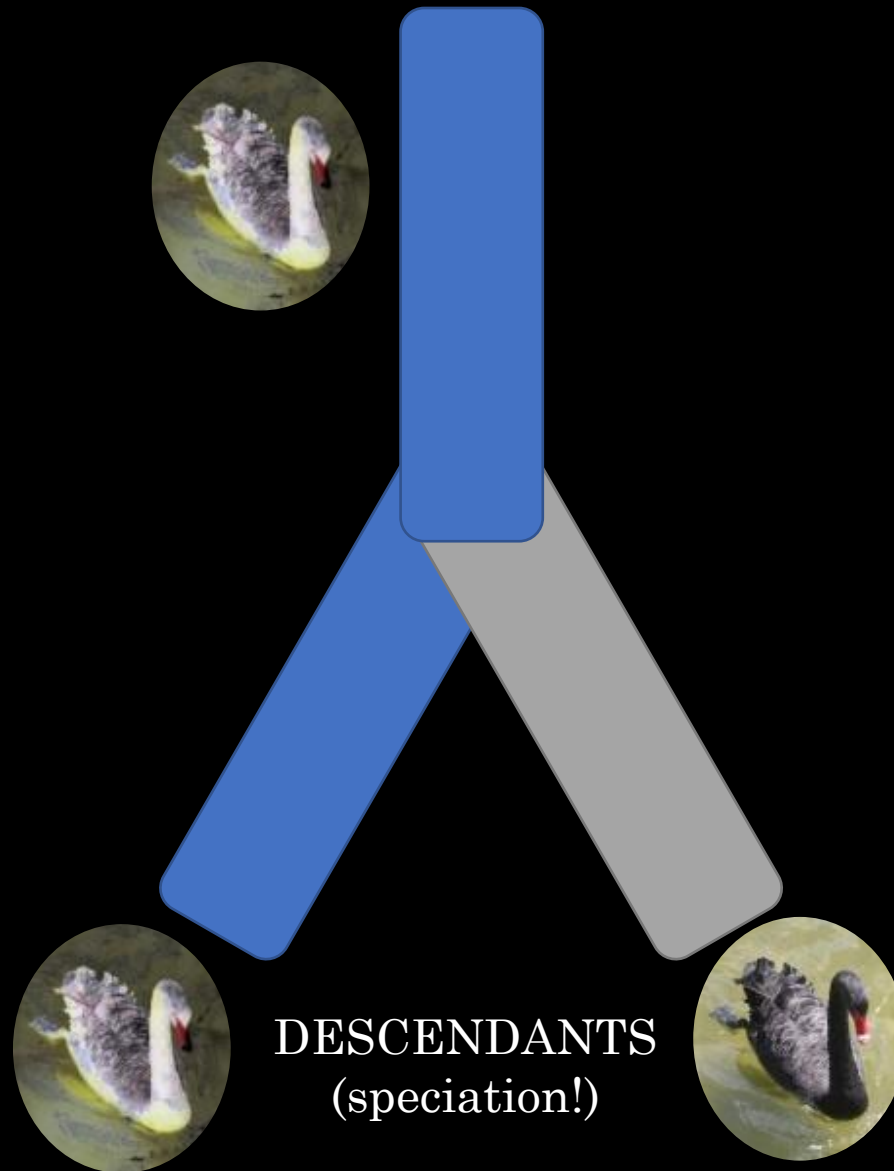
# dN/dS ratio comparison in short vs long RDV treatment



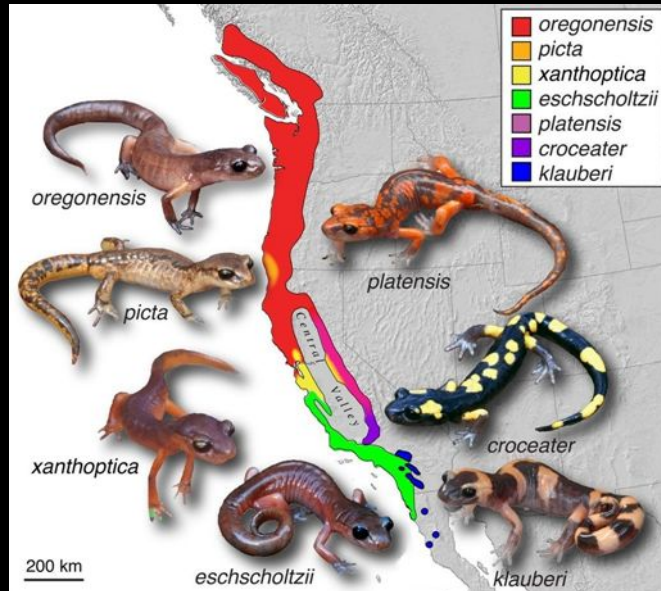
# Boundaries/migration also important



ANCESTOR

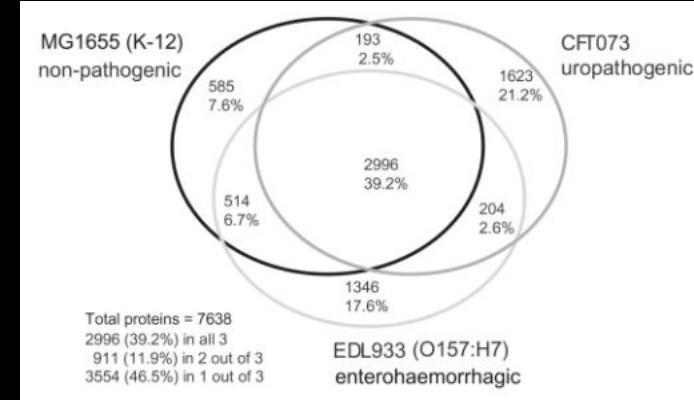


# Speciation ain't pretty

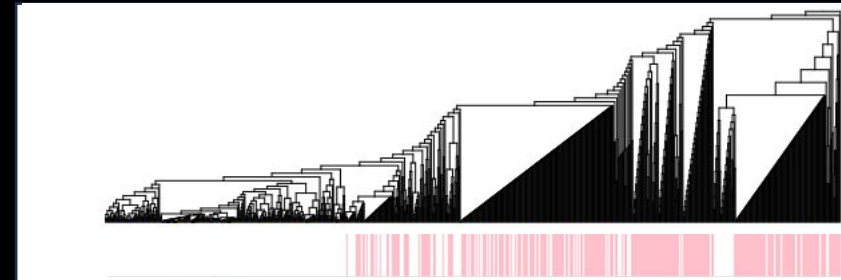


Ring species: most neighbours can interbreed, more-distant populations cannot

Devitt et al (2011) *BMC Evolutionary Biology*  
Welch et al (2002) *PNAS*  
Sanderson, Gray, et al. (2022)



Fewer than half of genes in these *E. coli* are found in all three of them



The erythromycin resistance gene *ErmB* is found in a scattered subset of *Enterococcus* genomes  
(pink = present in genome)

# Relationships among sequences

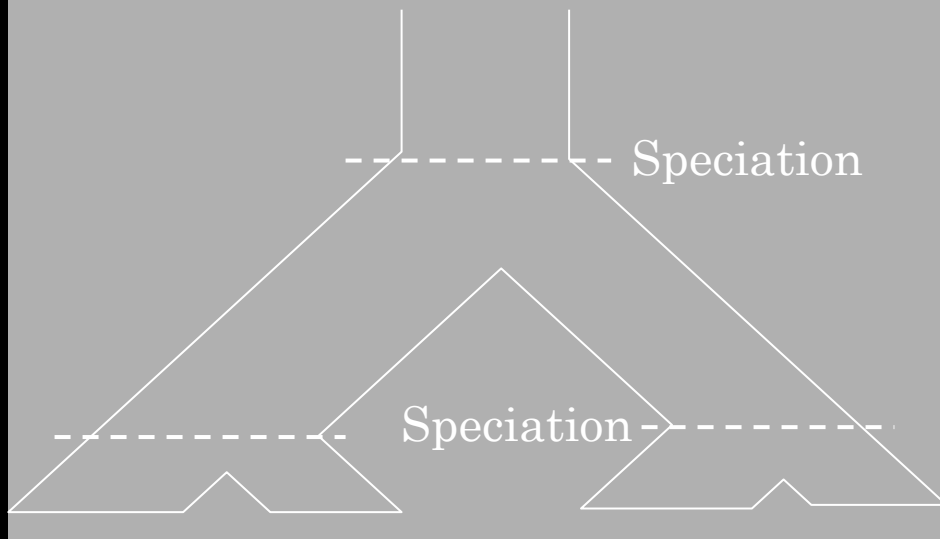
- HOMOLOGY
- ORTHOLOGY
- PARALOGY

Remember: duplications

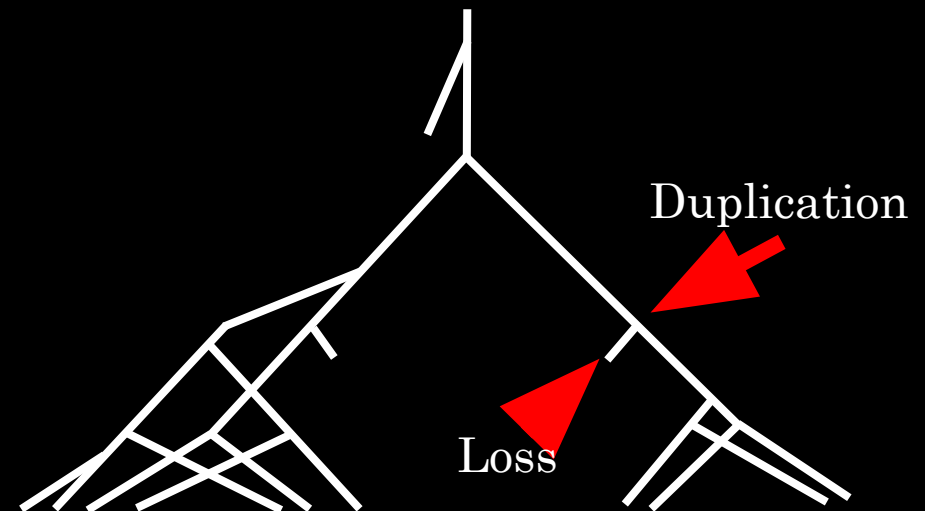
AAA ACC **GAA TCA** CCG GAT → AAA ACC **GAA TCG AAT CA** CCG GAT

# Bifurcations at the gene and species level

When speciation occurs, most genes are inherited by both descendants



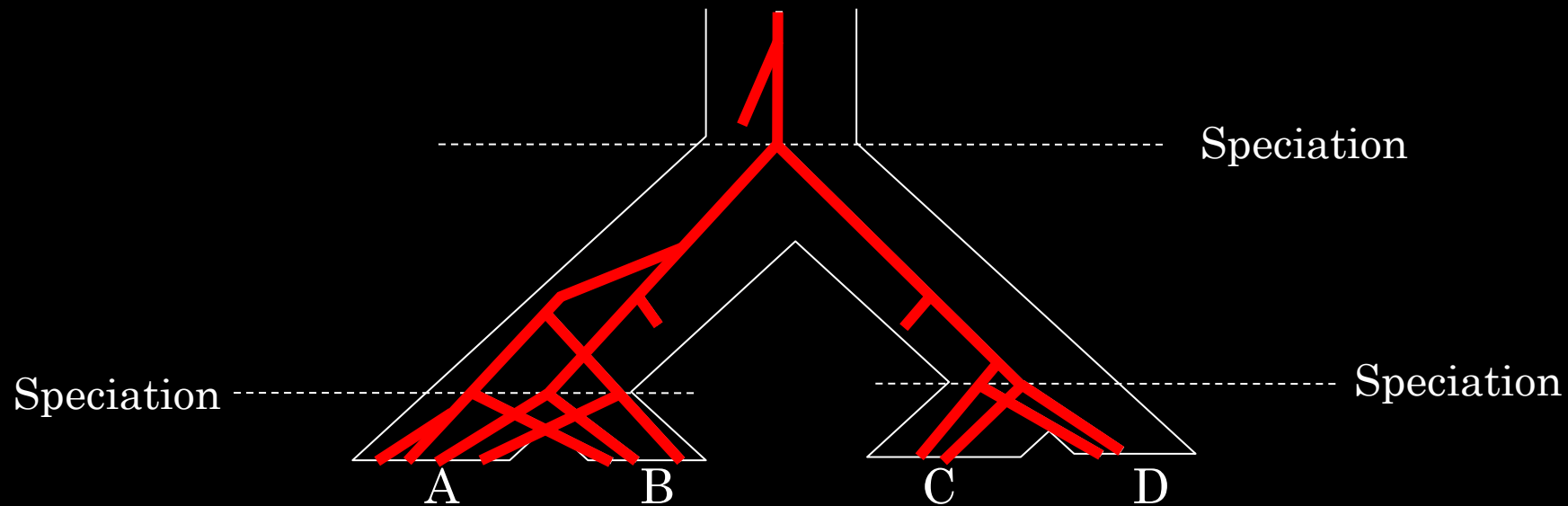
Genes can duplicate and disappear at any time, independently of speciation





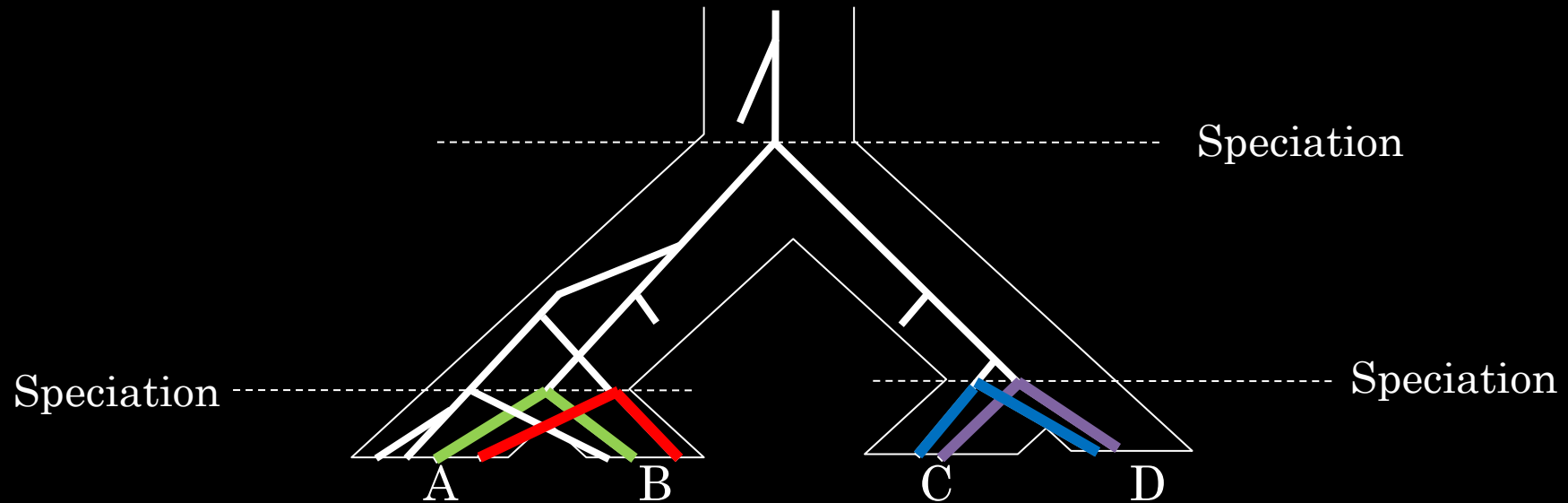
# Homology

- Any set of sequences (or, more generally, traits) that are descended from a common ancestor



# Orthology

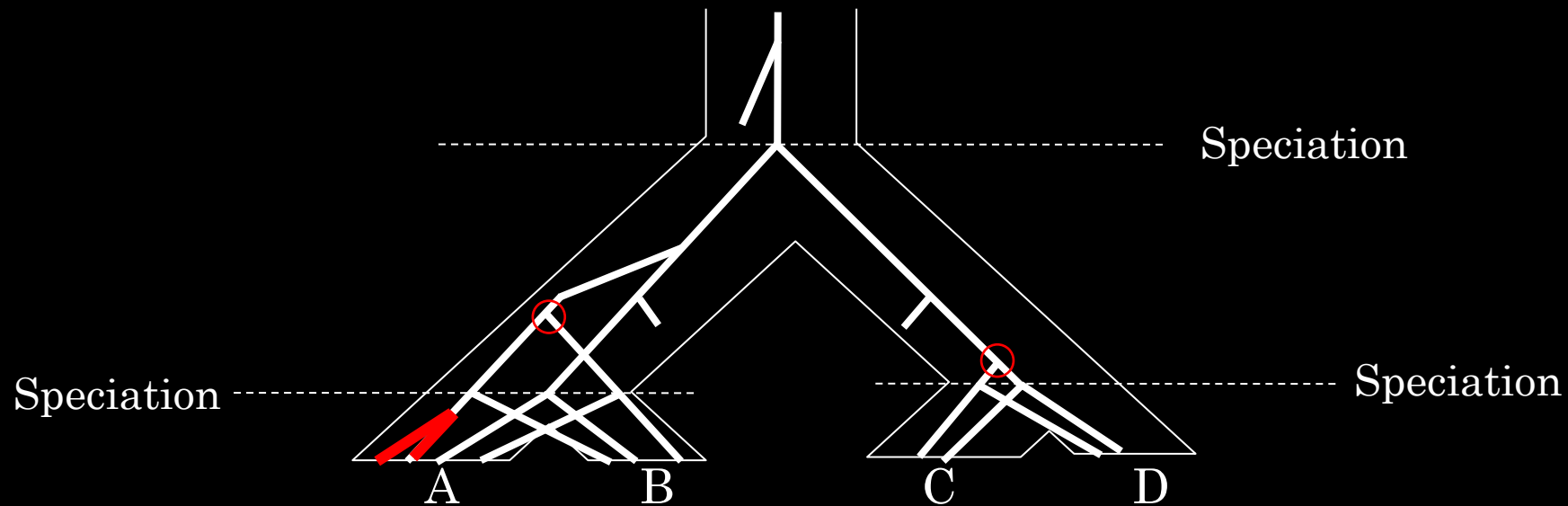
- Genes whose last common ancestor occurred at a speciation event





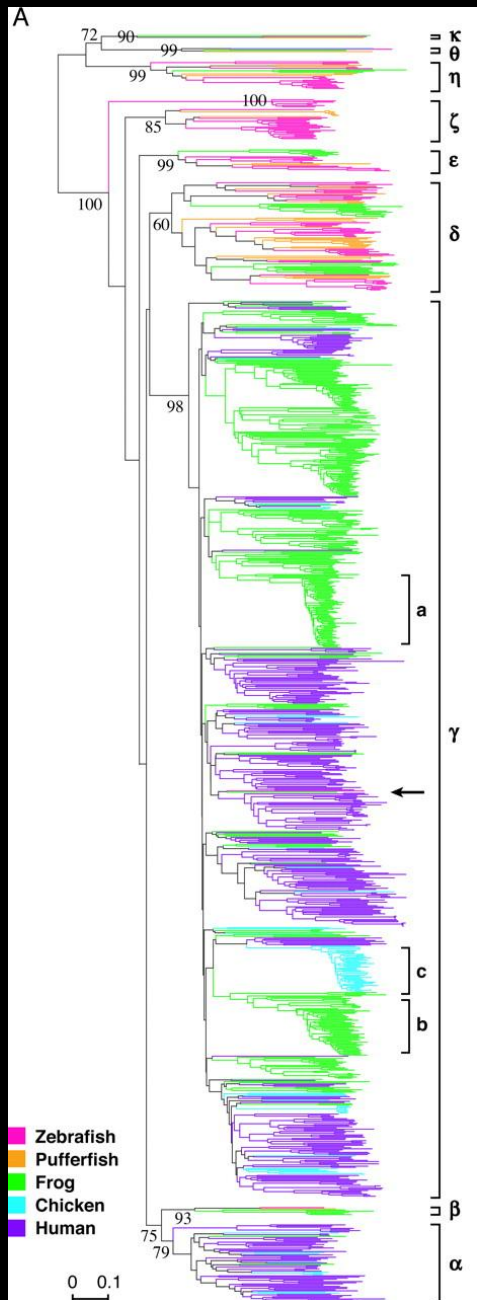
# Paralogy

- Two genes whose last common ancestor arose via a gene duplication event



# Why do we care about the difference?

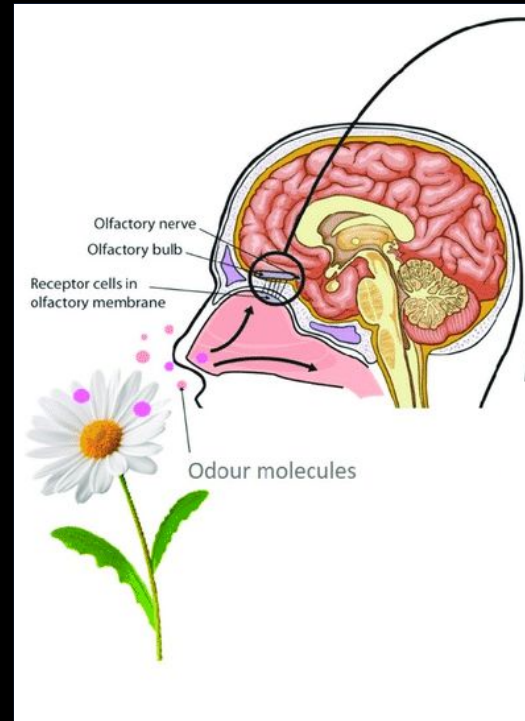
- The “ortholog conjecture”
- Orthologs are (kinda) the “same gene” in different species, so they are more likely to have evolved in similar ways and retained the same or similar function
- Paralogs are descendants in a single lineage – they are often lost, but since someone is still doing the original job, they may mutate to specialize over time, or “explore” new functions
  - Some antibiotic-resistance genes are paralogs of much more “innocent” genes



# Olfactory receptors

Many different target molecules, but all ORs:

- Sit in the cell membrane
- Recognize a specific set of odor molecules
- Trigger neural signalling

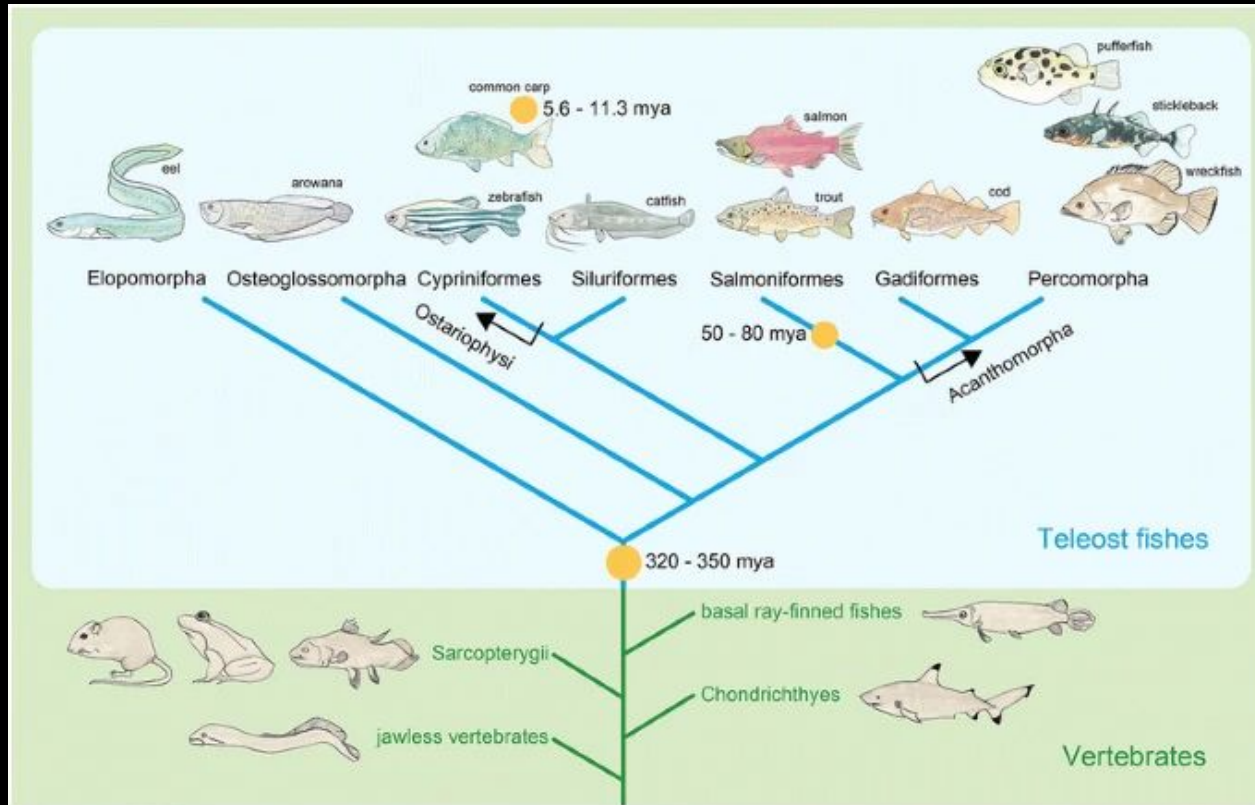


These sequences are all homologous, but trying to disentangle paralogy and orthology is a mess

Niimura and Nei (2005) *PNAS*

Vilela et al (2019) *Foods*

# Whole-genome duplication – mega paralogy!

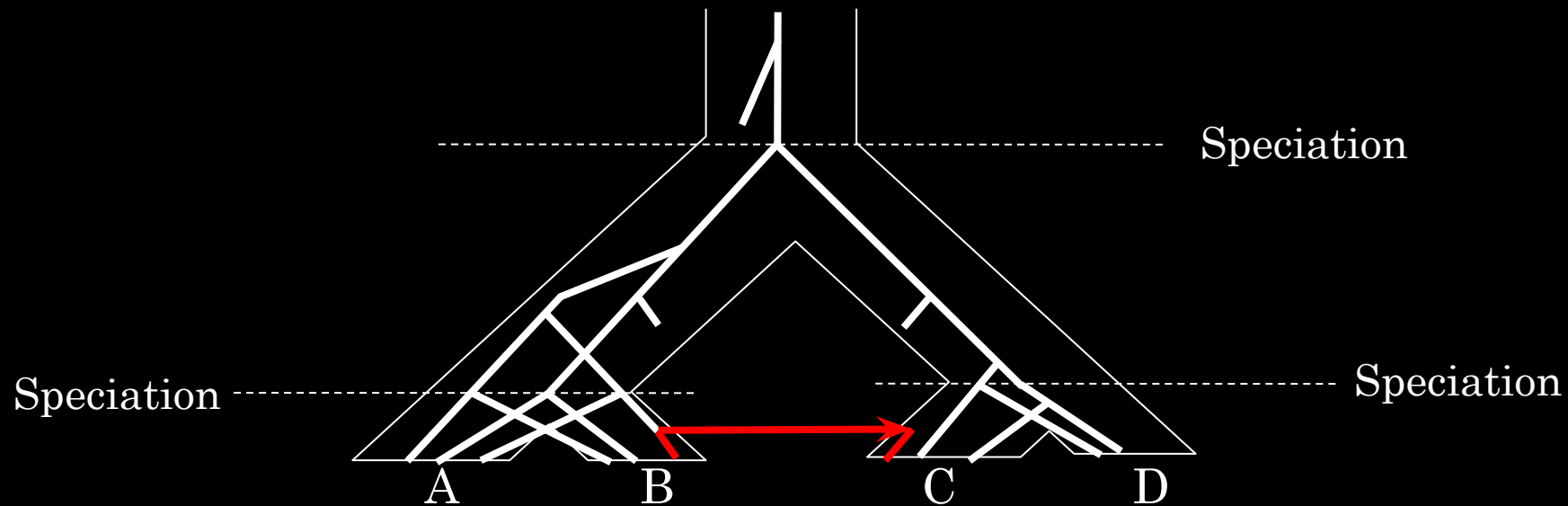


Descendants have 2x the genetic material of the ancestor

Many of the newly duplicated genes are lost or mutate and gain new / specialized functions

# Xenolog: lateral gene transfer

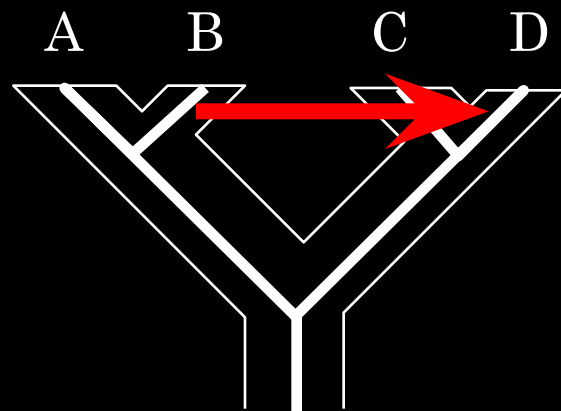
- Homologous sequences found in different species because of horizontal/lateral gene transfer



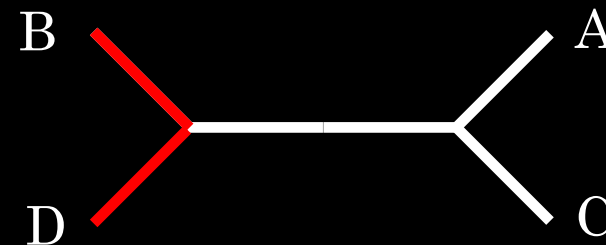
Uh-oh that ain't a tree anymore!

# Lateral gene transfer

- The transmission of genetic material between genomes in a manner other than parent to offspring
- A gene from organism 'B' is acquired by organism 'D' - thus B and D appear to be very closely related

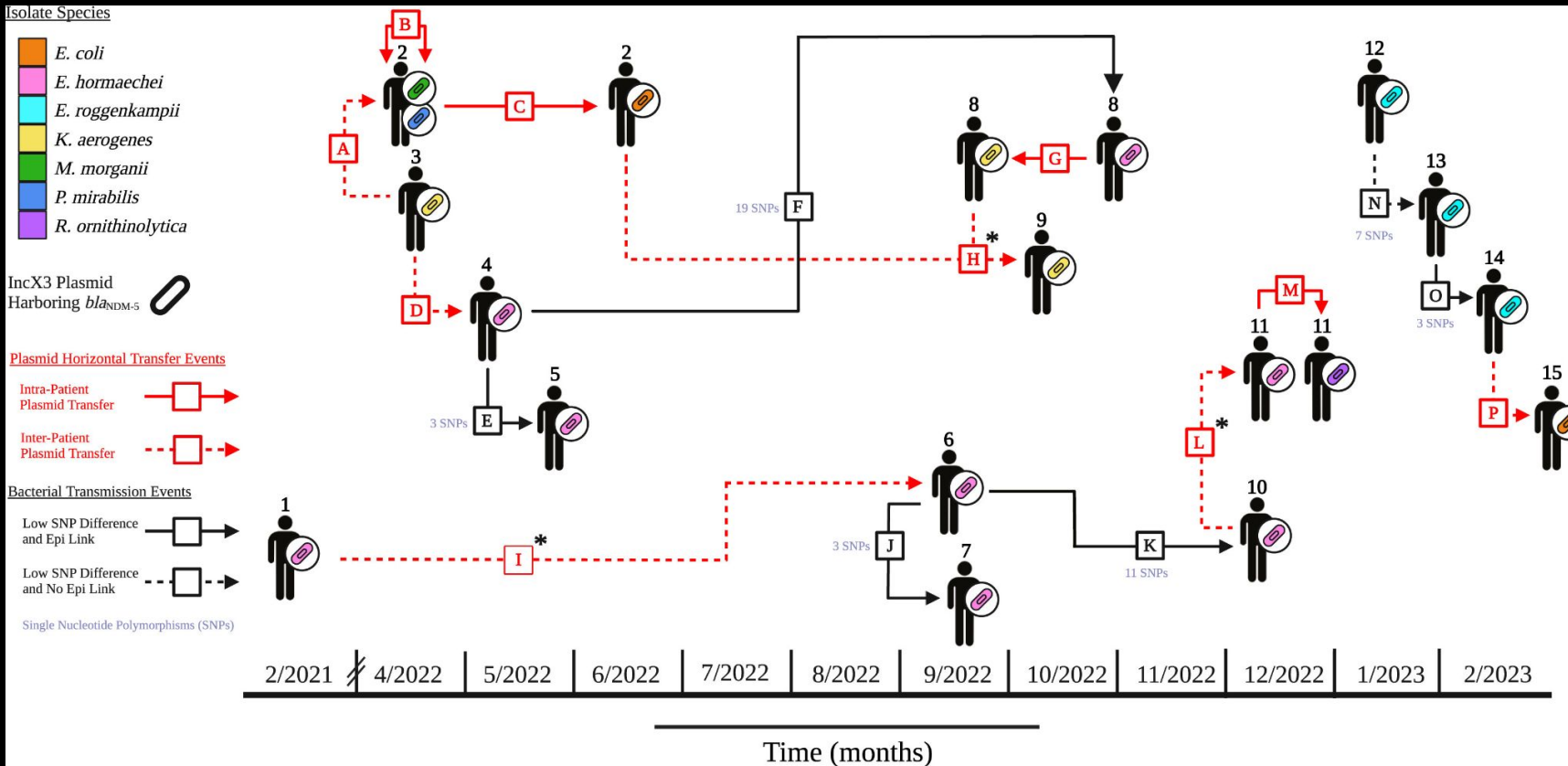


Organism phylogeny  
with LGT event

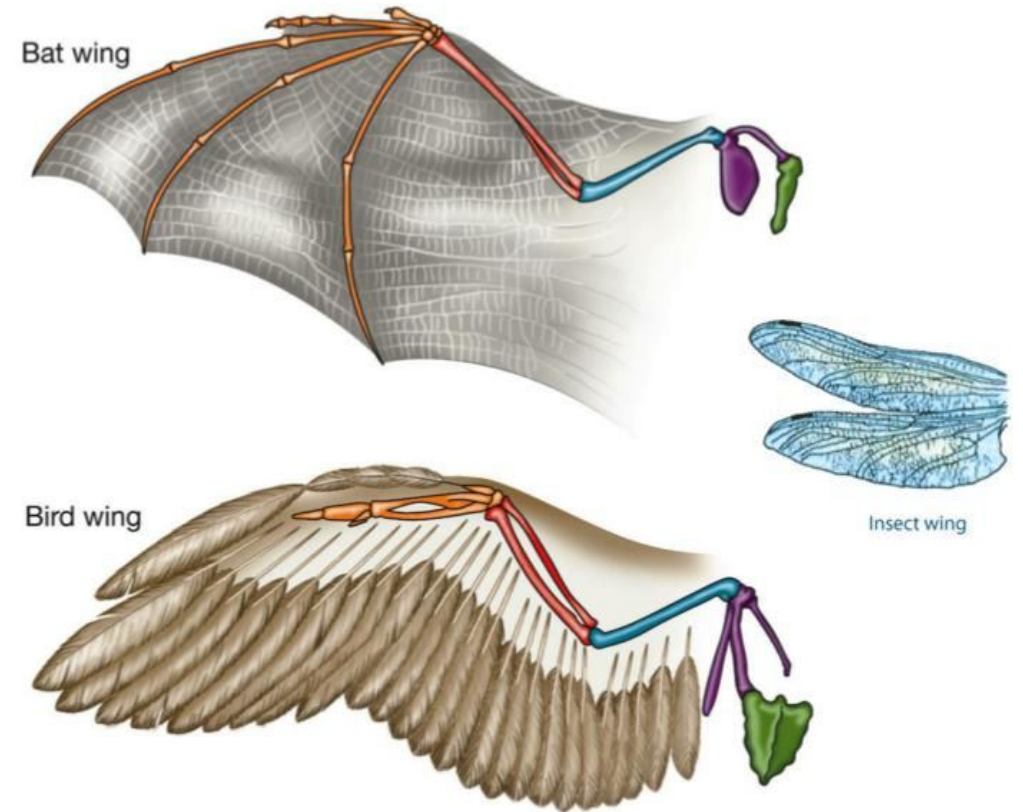
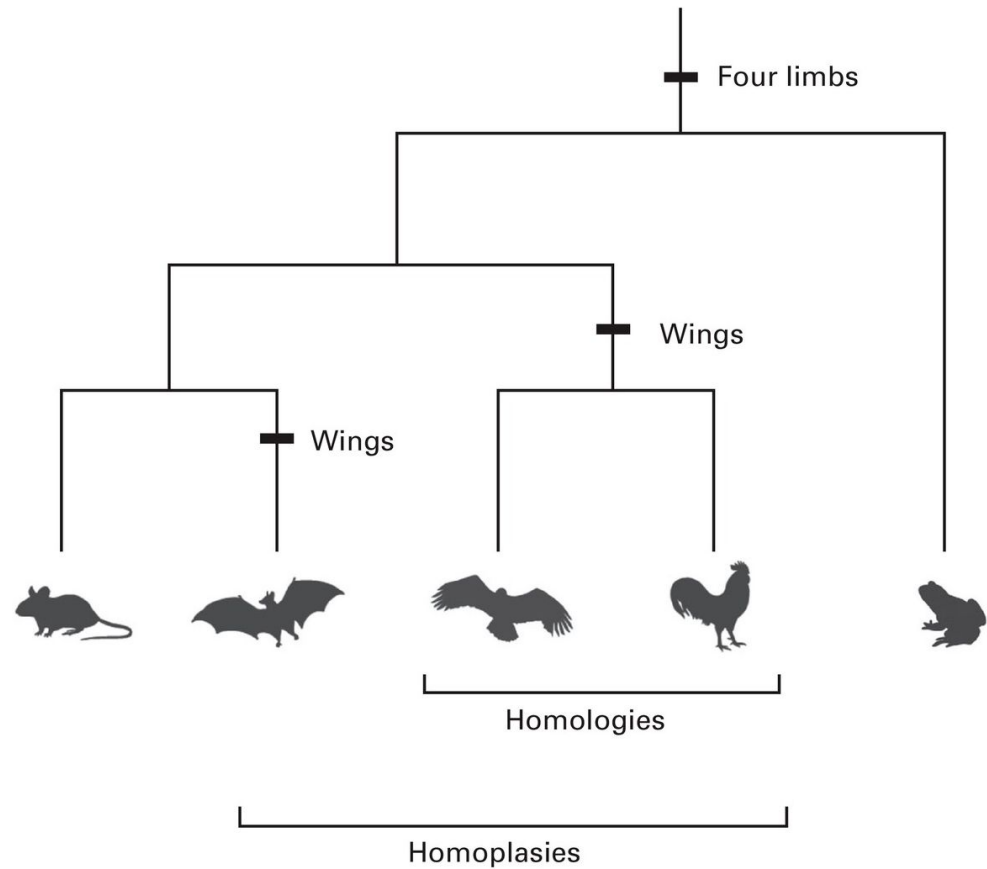


Inferred gene  
tree

# LGT has major clinical implications

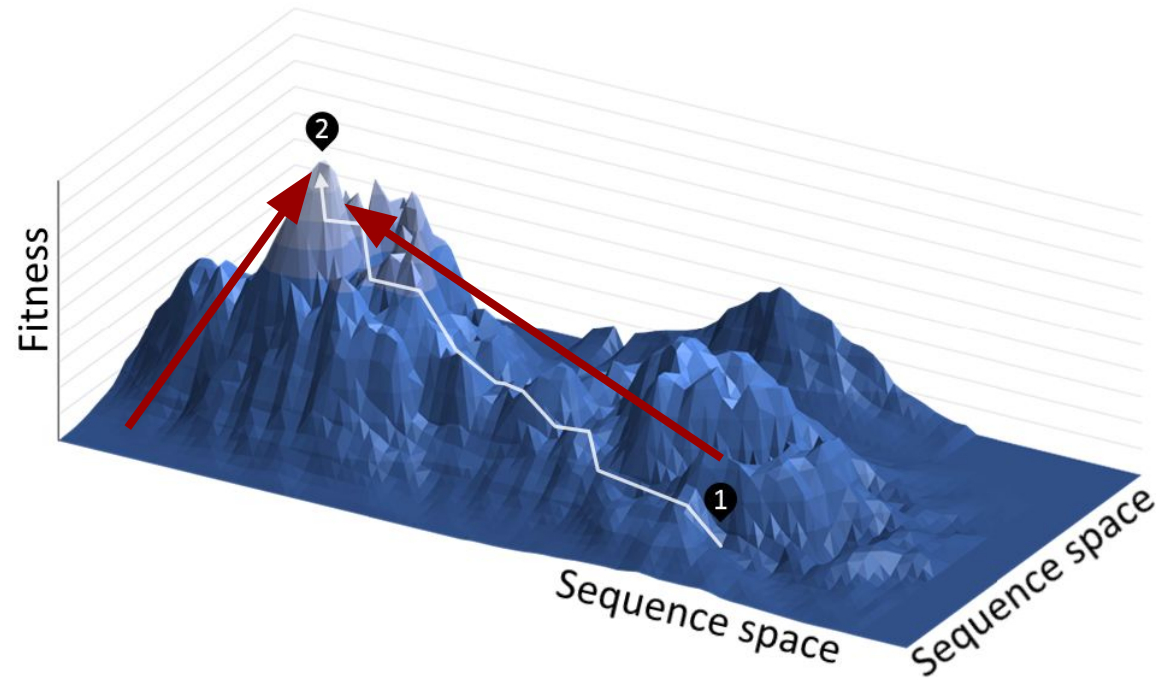
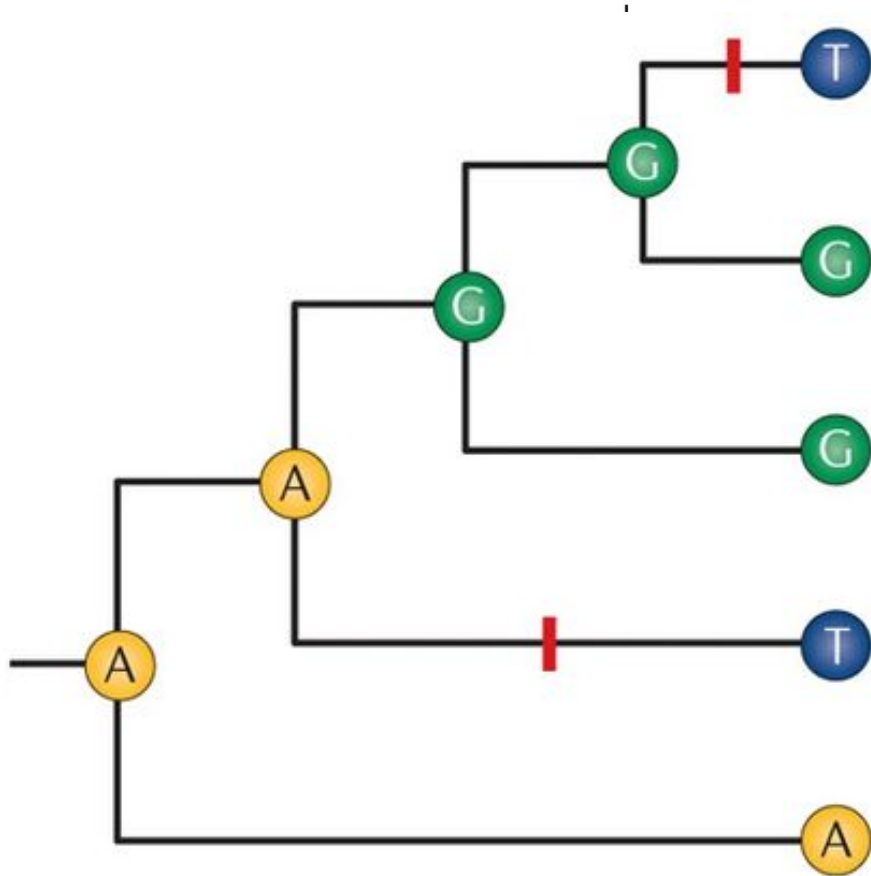


# Convergent evolution of wings

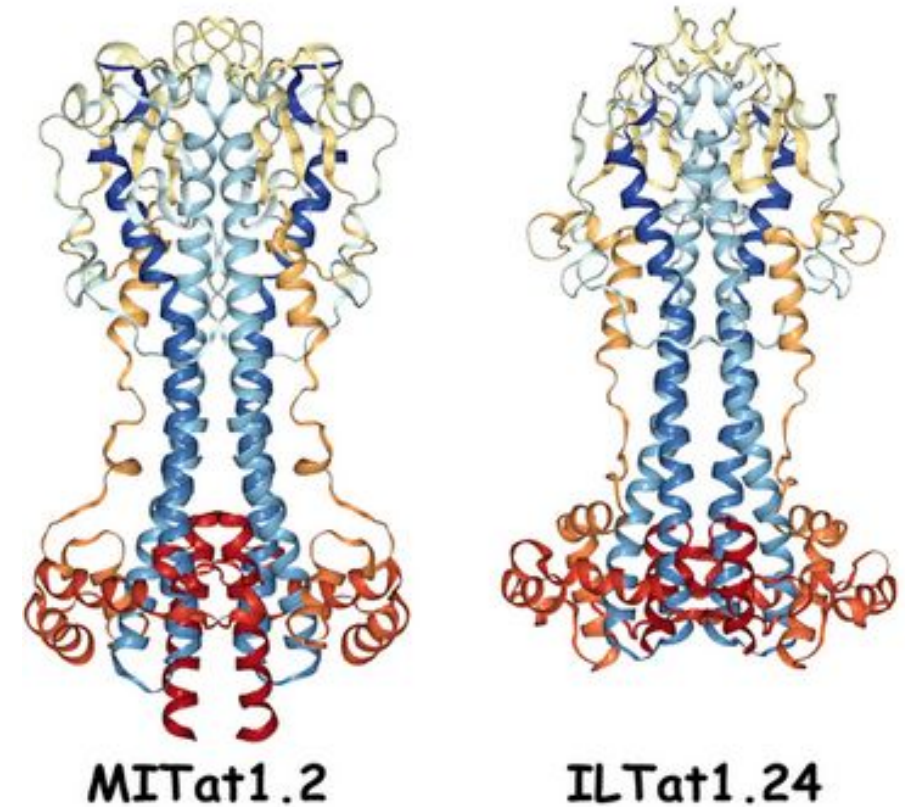
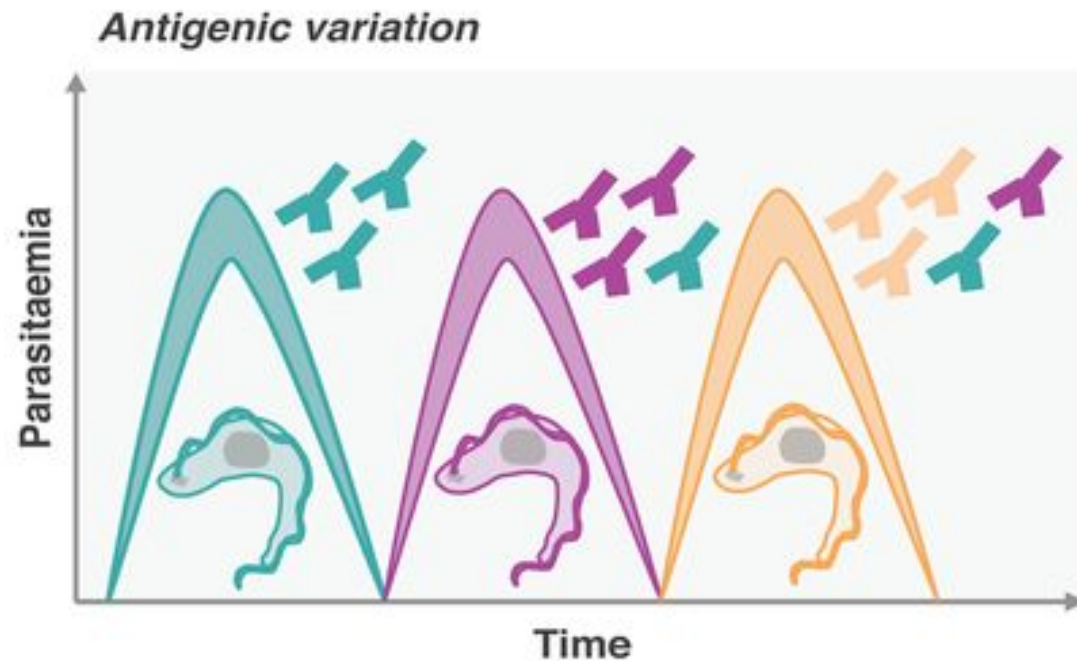




# Convergent (or homoplasious) mutations common



# Convergent structures without convergent sequences



16% sequence identity!

# Summary

Mutations change the gene sequence, and often the protein sequence as well

*Selection* weeds out less-effective versions of the protein (and thus the mutation)

By tracing the history of mutation and selection, we can recover the evolutionary relationships among biological sequences

# Summary

All genes that are related by descent from a common ancestor are **homologous**

Orthologs are sometimes thought of as ‘the same gene in different species’

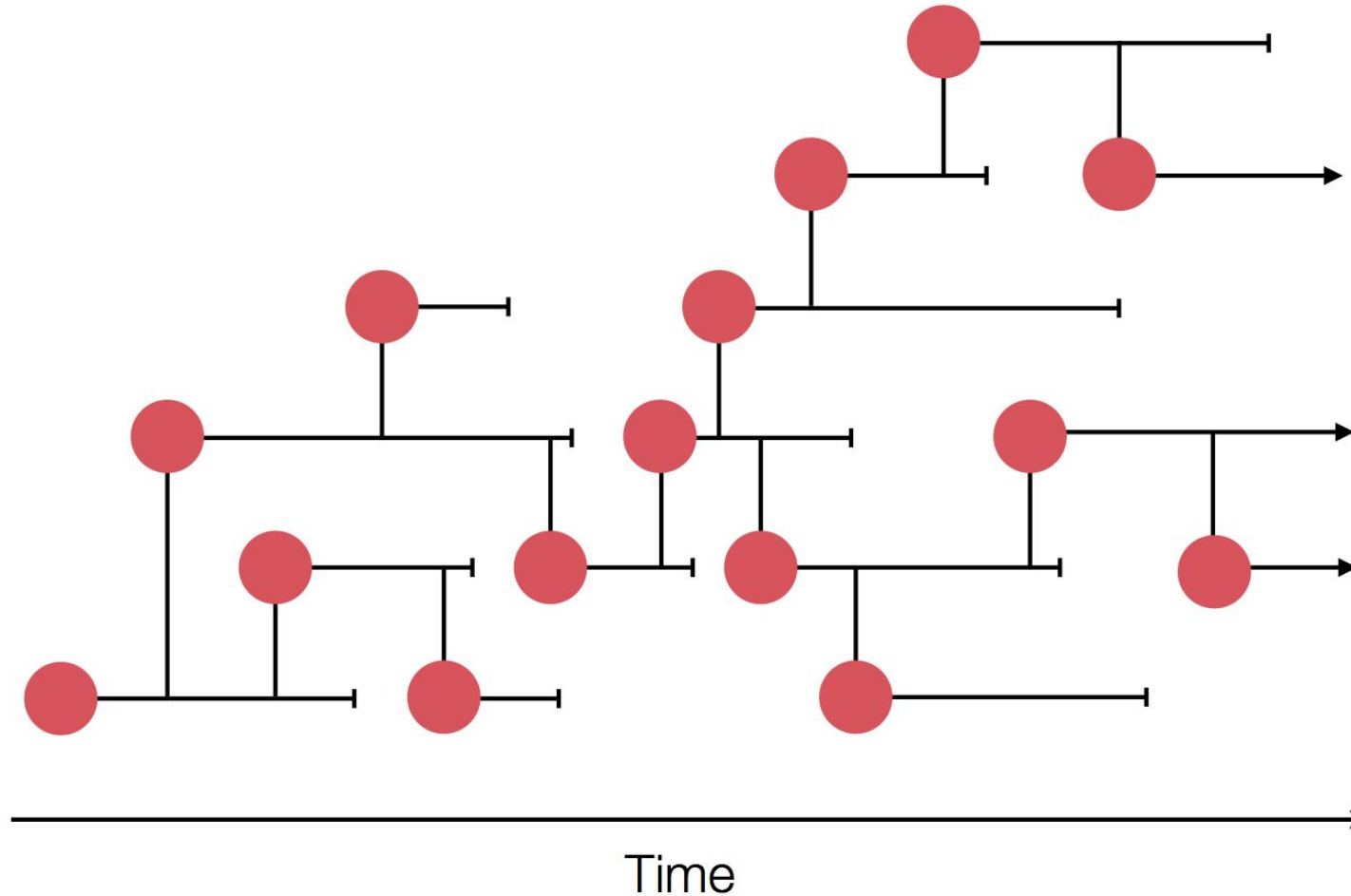
Paralogs arise by duplication, and can lead to proteins with new functions



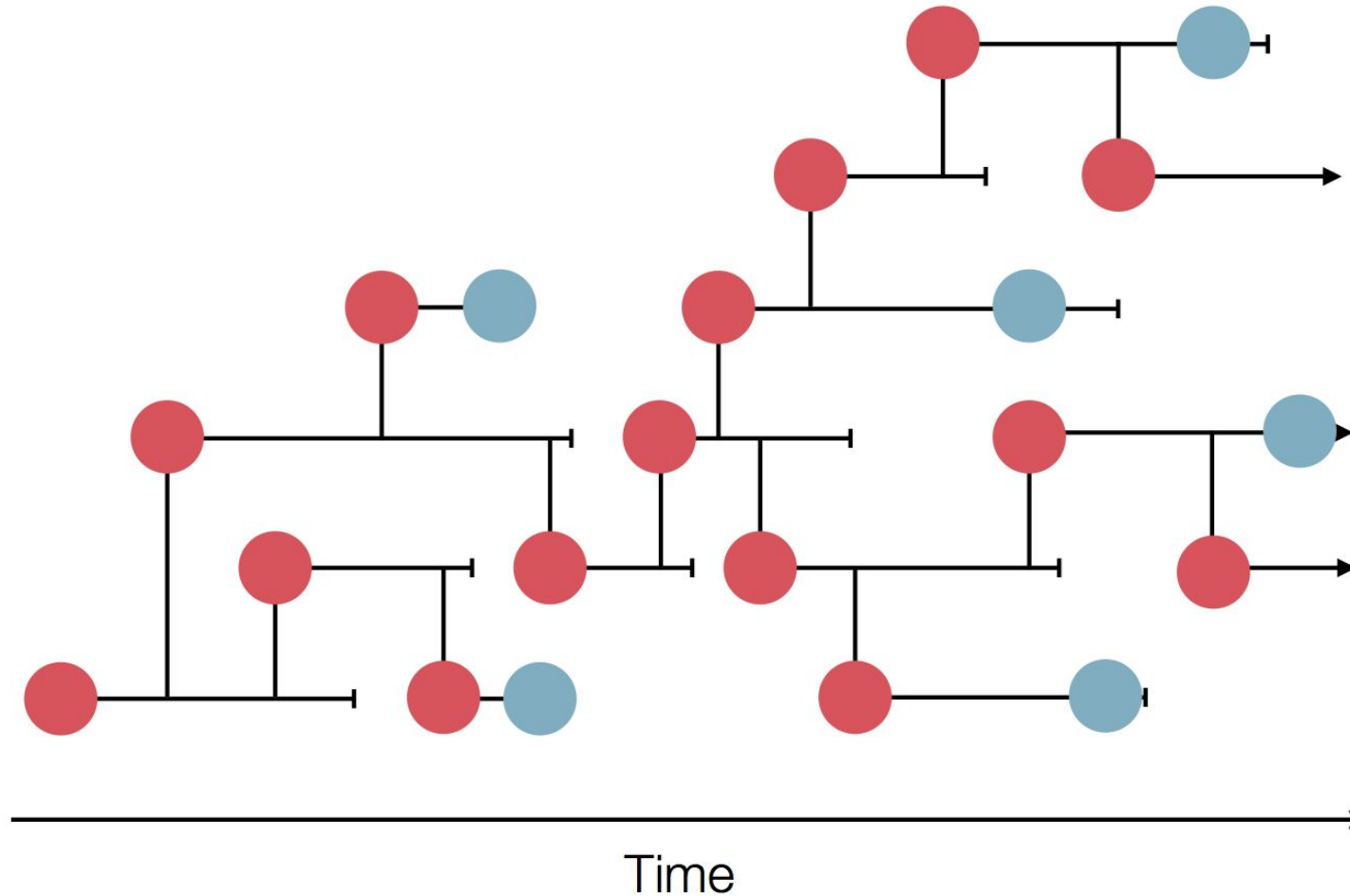




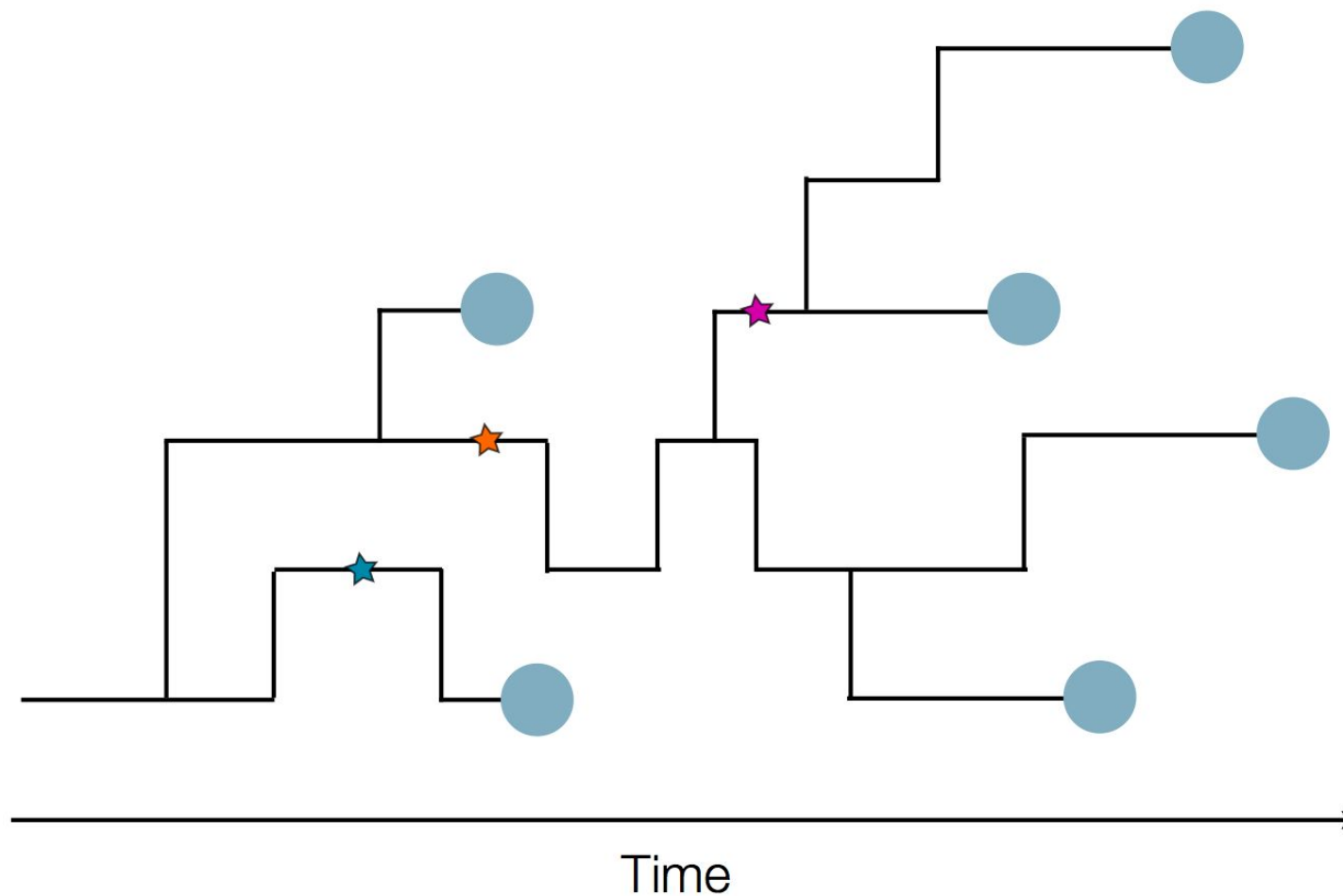
# Sampling & partially reconstructing underlying epidemic process



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